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The stratoid granites of central Madagascar: paleomagnetism and further age constraints on neoproterozoic deformation

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Abstract

A paleomagnetic and $^{40}Ar/^{39}Ar$ study of a 630-Ma alkaline granite suite in Madagascar, the so-called 'stratoid' granites, reveals a complex history of remagnetization during the formation of the Antananarivo *Zone de Virgation* at ~560 Ma (D_2) and the Angavo shear zone at ~550 Ma (D_3). $^{40}Ar/^{39}Ar$ dating of hornblende, biotite and potassium feldspar from rocks affected by D_2/D_3 show initial cooling rates of 8 °C/Ma during the 550–520 Ma interval followed by slower cooling of 2.5 °C/Ma. The thermal effects of the D_2 and D_3 events appear to be restricted to regions surrounding the shear zones as evidenced by a $^{40}Ar/^{39}Ar$ biotite age of 611.9±1.7 Ma north of the virgation zone. The paleomagnetic data from the stratoid granites are complex and some sites, particularly in areas to the north of the virgation zone, may have been rotated about non-vertical axes following their emplacement and cooling. Because of these possible rotations, our best estimate for the paleomagnetic pole for Madagascar is derived from sites within the virgation zone. This pole falls at 6.7°S, 352.6°E ($a_{95} = 14.2°$). A post-metamorphic cooling history for the virgation zone indicates a magnetization age of 521.4±11.9 Ma. Our work in central Madagascar, coupled with previous studies, suggests that emplacement of the 630 Ma stratoid granites followed a collisional (?) tectonic event beginning around 650 Ma, recently recognized in southern Madagascar and in Tanzania. Subsequently, the stratoid granites in the Antananarivo virgation zone were reheated (~750–800 °C) at pressures between 3.5 and 3.6 kbars resulting in a pervasive remagnetization. We suggest that the younger shear events are genetically related to collisional tectonics elsewhere during the final stages of Gondwana assembly and are a consequence of the Kuunga Orogeny further south.
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Keywords: Madagascar; East African Orogen; Paleomagnetism; $^{40}Ar/^{39}Ar$ dating; Kuunga Orogen

1. Introduction

The island of Madagascar is located in a key position within the assembled Gondwana supercontinent (Fig. 1). Its location prior to the assembly of Gondwana and its exact relationship

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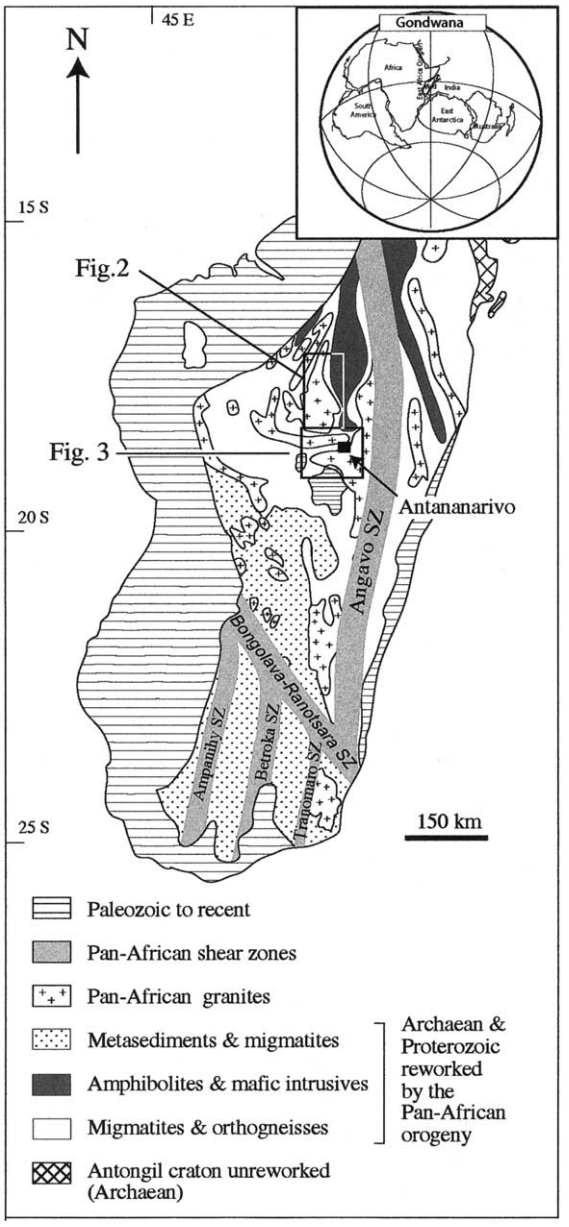


Fig. 1. Generalized geologic of Madagascar showing the major neoproterozoic shear zones and the locations of Maps 2 and 3.

to both East and West Gondwana has been the subject of considerable debate. In part, this is due to a lack of early Palaeozoic and Neoproterozoic paleomagnetic data from the island. Recent work (Meert et al., 2001b) provided the first lower Palaeozoic (508 Ma) paleomagnetic pole from Madagascar. In addition to that paleomagnetic work, geochronologic investigations of Madagascar during the past 10 years have provided important constraints on the evolution of the island (Paquette et al., 1994; Kröner et al., 1996; Ito et al., 1997; Paquette and Nédélec, 1998; Handke et al., 1999; Tucker et al., 1999a,b; Kröner et al., 2000; De Wit et al., 2001). Summaries of the geochronologic data and their tectonic significance are provided by Kröner et al. (2000), Meert (in press). Detailed structural studies were performed in the southern part of the island by Martelat et al. (1997), Martelat et al. (2000), De Wit et al. (2001), and in the northern-central parts of the island by Nédélec et al. (1994, 2000), Grégoire (1999), Collins et al. (2000); however, the geodynamic significance of the main magmatic and metamorphic events are still a matter of debate (Collins and Windley, 2002).

Widespread magmatic activity in central Madagascar (gabbroic and granitoid intrusions) is dated between 740 and 824 Ma (Handke et al., 1999; Tucker et al., 1999a,b; Kröner et al., 2000). Handke et al. (1999) concluded that the magmatism resulted from Andean-type subduction of the Mozambique Ocean beneath central Madagascar. Slightly younger ages (715–754 Ma, U–Pb) are also found within the metamorphosed Daraina sequence in the northernmost part of Madagascar (Tucker et al., 1999a,b) and these rocks are in turn correlated with similar-age magmatic rocks in the Seychelles and also the Rajasthan region of India. Ashwal et al. (2002), Tucker et al. (2001) suggest that the observed magmatism in northwest India, central, northern Madagascar and the Seychelles between 700 and 800 Ma formed in an active margin setting near the periphery of East Gondwana. Kröner et al. (2000) are less supportive of an arc setting for these rocks and favour a position of Madagascar either adjacent to East Africa or as a separate (from India) block within the Mozambique Ocean. They suggest, in addition to arc-magmatism, a possibility for magmatic underplating in another tectonic setting as the source for these magmatic rocks. The latter is consistent with the results of Guerrot et al. (1993) on the geochemical nature of the mafic intrusives from the 780 Ma Andriamena Complex north of

Antananarivo, suggesting an intracontinental rift setting. Whatever the actual tectonic setting, there is a broad consensus that the large magmatic production in central and northern Madagascar between 715 and 824 Ma is contemporaneous with the rifting and dispersal of Rodinian fragments.

The next important magmatic event occurred around 630 Ma and corresponds to the emplacement of alkaline 'stratoid' granites in northern-central Madagascar (Nédélec et al., 1995; Paquette and Nédélec, 1998) and to the emplacement of anorthosite bodies in southern Madagascar (Ashwal et al., 1998, 1999). The so-called 'stratoid' granites (Figs. 2 and 3) were recognized as a conspicuous geologic feature in central Madagascar by the early works of Guyonnaud (1951), Besairie (1968). They outcrop in an 800 km long north–south trending belt and consist mostly of conformable granitic sheets ranging in thickness from 0.5 to more than 500 m within gneissic and migmatitic host rocks. Over the whole of central Madagascar, these granites are characterized by a N–S striking foliation gently dipping to the west and a shallowly plunging mineral lineation trending west–southwest (Nédélec et al., 1994; Grégoire, 1999). The granites were emplaced at mid-crustal levels (4–5 kbars) in a region of elevated heat flux (Nédélec et al., 1994; Grégoire, 1999). The tectono-magmatic event responsible for their syntectonic emplacement is referred to as D_1 by Nédélec et al. (2000). According to Paquette and Nédélec (1998), Meert (in press), this event may represent post-collisional extension at 630 Ma following the initial collision of Madagascar with East Africa at ~650 Ma. The timing of collisional orogenesis in Madagascar is consistent with recent dating in Tanzania by Muhongo et al. (2001). To

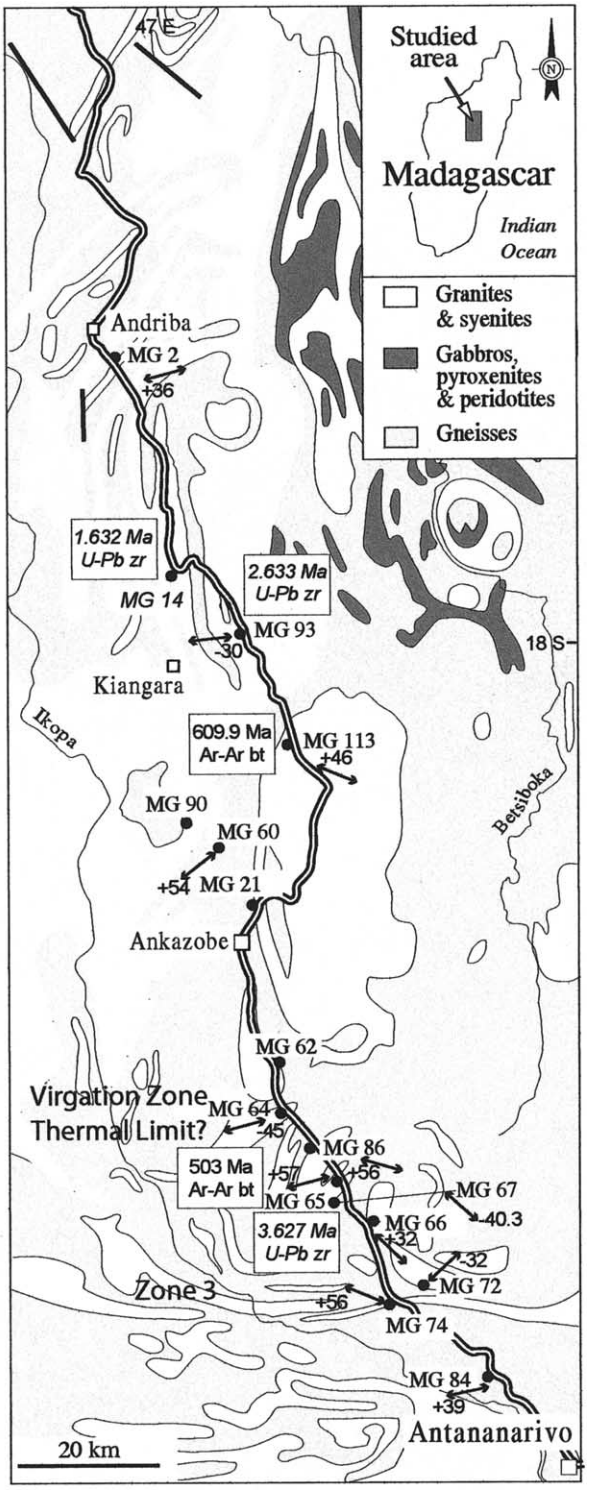


Fig. 2

Fig. 2. Map of the northern segment of the study area. Individual sampling sites are shown along with an indication of the paleomagnetic declination and inclination (arrows indicate declination, numbers indicate inclination). The locations of sampling sites and age determinations are also shown in this figure. U–Pb Zircon ages are from Paquette and Nédélec (1998) as follows: (1) 632±2 Ma (MG-14); (2) 633±1 Ma (MG-93) and (3) 627±2 Ma (MG-65). Zone 3 refers to the 'North Virgation' structural domain of Nédélec et al. (2000). $^{40}Ar/^{39}Ar$ ages/sites listed in Table 3.

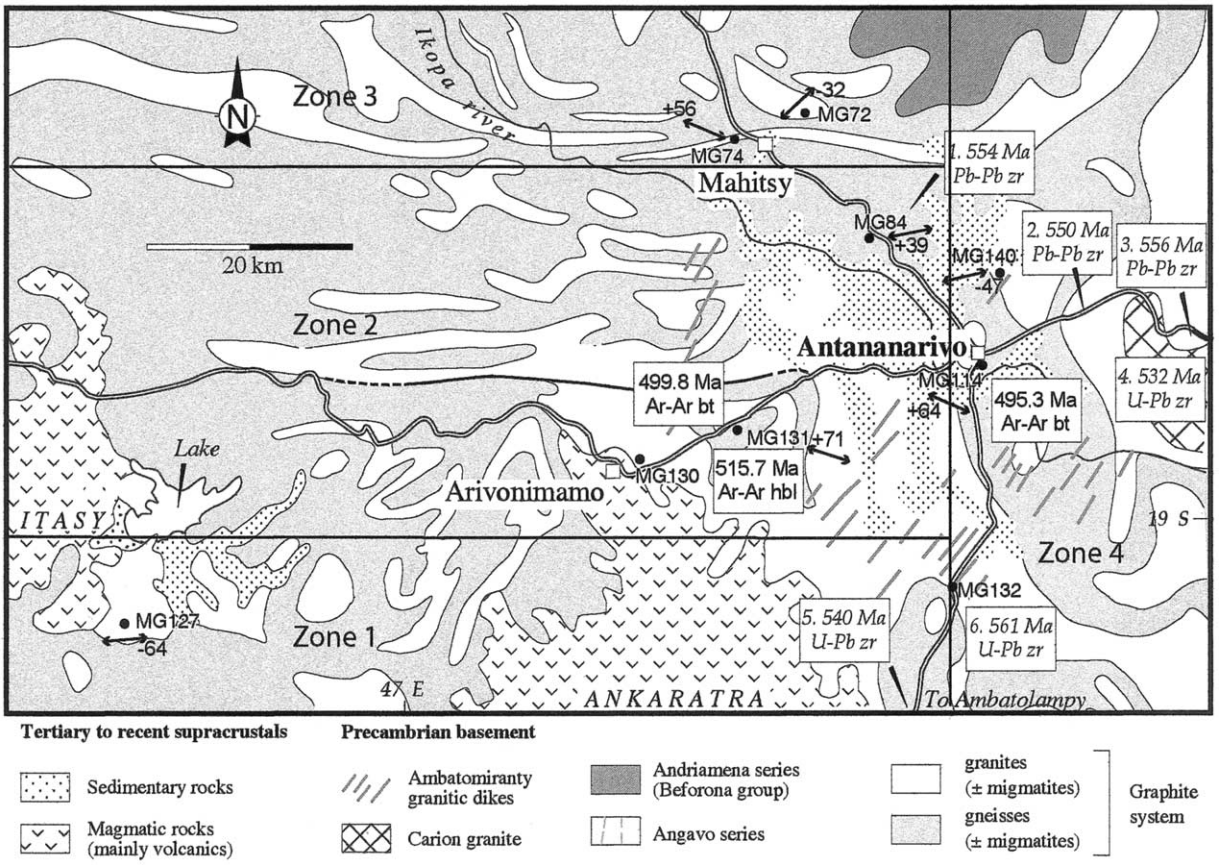


Fig. 3. Map of the Antananarivo region. Individual sampling sites are shown along with an indication of the paleomagnetic declination and inclination (arrows indicate declination, numbers indicate inclination). The locations of geochronologic sites in this study and previous studies of Paquette and Nédélec (1998), Meert et al. (2001a), Tucker et al. (1999a), Kröner et al. (1999, 2000) are shown as follows: (1) 554.4±1 Ma (MAD-79 of Kröner et al., 2000); (2) 550.1±1.4 Ma (MAD-50 of Kröner et al., 2000); (3) 556.0±1.7 Ma (MAD-54 of Kröner et al., 1999) Note: sample MAD-54 was incorrectly identified by the authors as part of the Carion granite; (4) 532.1±5.2 Ma (Carion granite of Meert et al., 2001a); (5) 540±3 Ma (undeformed pink granite LA-300A of Tucker et al., 1999b) and (6) 561±4 Ma (MG-132 Ambatomiranty dike of Paquette and Nédélec (1998). The region is divided into structural domains as described in Nédélec et al. (2000) as follows: Zone 1: 'West Ambatolampy Domain'; Zone 2: 'South Virgation'; Zone 3: 'North Virgation' and Zone 4: 'West Angavo'. $^{40}Ar/^{39}Ar$ ages/sites listed in Table 3.

the south, De Wit et al. (2001) obtained conventional U–Pb zircon and monazite ages in the 627–647 Ma range, which they assigned to a first thrusting event coeval with the emplacement of syntectonic anorthosites hence supporting the notion that collisional tectonism in Madagascar occurred during that same interval. Therefore, we conclude that Madagascar experienced a strong tectonothermal imprint between ~620 and 650 Ma although the extent and significance of this event are debated.

Finally, it is clear that much of the island of Madagascar has experienced thermal overprinting and structural reworking due to motion along Late Neoproterozoic shear zones (Rolin, 1991; Martelat et al., 1997; Nédélec et al., 2000; De Wit et al., 2001). These major shear zones are shown in Fig. 1. Nédélec et al. (2000) identified two such younger deformational phases in the region of Antananarivo. The E–W trending Antananarivo *Zone de Virgation* (French, 'flexural zone'; hereafter referred to as the virgation zone) resulted from

sinistral transpression under high-grade conditions (4–5 kbars, 700–800 °C) and was assigned D_2. The age of D_2 is constrained to $\geqslant 561$ Ma by U–Pb dating of the Ambatomiranty dykes that sealed the virgation (Fig. 3, Paquette and Nédélec, 1998). To the east, the N–S Angavo shear zone is a major feature of eastern Madagascar, likely 700 km in length and 40 km in width. The Angavo shear zone was first recognized east of Antananarivo by Ralison and Nédélec (1997) and then, 400 km further south, near Ifanadiana by Martelat et al. (2000). Cessation of activity in the Angavo shear zone and the Antananarivo virgation zone is constrained by SHRIMP dating of the late-syn to post-D_3 Carion granite at 532.1 ± 5.2 Ma (Fig. 3, Meert et al., 2001b). In an effort to constrain the paleogeography of Madagascar during the 630–500 Ma interval, we undertook a detailed paleomagnetic and $^{40}Ar/^{39}Ar$ study of the stratoid granites in central and northern Madagascar (i.e. the rocks that either underwent the D_2 virgation effect or escaped this rotation).

2. Sampling

Figs. 2 and 3 show the distribution of paleomagnetic sampling sites in the stratoid granites along with the locations of U–Pb and $^{40}Ar/^{39}Ar$ ages. We collected 140 samples from 20 sites within the stratoid granites. Eleven sites (MG 2, 21, 60, 62, 64, 65, 67, 86, 90, 93 and 113: Fig. 2, and MG 127 in Fig. 3) belong to regions identified by Nédélec et al. (2000) as structurally unmodified by D_2 or D_3. The remaining sites correspond to rocks whose structures were rotated by the D_2 virgation (MG 66, 72, 74, 84, 114, 131 and 140a; Fig. 3) or to the Ambatomiranty dykes that sealed the D_2 virgation (MG 132).

Samples were drilled in the field using a gasoline-powered drill and oriented using both magnetic and sun compasses. The samples were then cut into specimens at Indiana State University and both bulk susceptibility and anisotropy of magnetic susceptibility were measured using a Sapphire Instruments SI-2B susceptibility meter. Individual samples were then stepwise treated using thermal (60 min at each temperature level) or alternating field demagnetisation and a treatment sequence was selected that resulted in the clearest separation of individual vector components. Mean susceptibilities averaged 2.8×10^{-2} SI (range $2.8 \times 10^{-3} - 7.0 \times 10^{-2}$ SI) and natural remanent magnetization intensities averaged 980 mA/m (range from 20 to 9700 mA/m).

3. Results

3.1. Paleomagnetic results

Table 1 lists the paleomagnetic results from the stratoid granites in this study. A number of samples exhibited chaotic behavior during demagnetisation and reached no stable endpoint direction and results from these samples were not included in the analyses (Table 1). Typical demagnetisation plots are shown in Figs. 4 and 5. At nine sites, a consistent overprint direction was removed at low temperatures and coercivities (Fig. 4c). The mean overprint direction has a declination = 342° and an inclination of −19.8° ($k = 23$, $a_{95} = 11.8°$; Fig. 6a, b and Table 2). This direction is slightly shallower than either the present dipole field for central Madagascar (Dec = 0°, Inc = −34°) or the present earth's field direction calculated for 1997 from the 1995 IGRF (Dec = 347°, Inc = −52°). However, it does lie along a great circle trajectory between the present dipole field and the characteristic high-temperature direction and likely results from a present-field overprint superimposed on the primary direction (see also Fig. 4a and Fig. 6b). The high temperature or coercivity direction was observed in 94 samples from 20 sites and showed dual polarities. Because three sites contained only one sample, we calculated mean directions based on sample means and site means (for sites with more than one sample). The normal polarity sample mean was Dec = 77.8°, Inc = −48.1° (k = 9.7, $a_{95} = 7.8°$; Fig. 6c and Table 2) and was nearly identical to the site mean direction of Dec = 83.0°, Inc = −46.0° ($k = 11.9$, $a_{95} = 20.3°$; Fig. 6d and Table 2). The reverse sample mean direction is calculated at Dec = 280.3°, Inc = +49.4° ($k = 9.5$, $a_{95} = 6.6°$; Fig. 6c and Table 2) also identical to the site mean direction of Dec = 276.6°, Inc = 53.9°

Table 1
Stratoid granite paleomagnetic results

Site number	Lat/Long	*n*/*N*	Declination	Inclination	Kappa (κ)	a_{95}	VGP latitude	VGP longitude	dp, dm
MG-2	17°39.5′S, 46°57.5′E	4/4	255.0	36.3	35	15.8	19.6S	332.7E	10.7, 18.4
MG-21 (LT)	18°16.8′S, 47°08.4′E	7/8	345.0	−12.0	11	19.4	70.9S	175.1E	10.0, 19.7
MG-21 (HT)		1/8	43.0	−57.0	–	–	47.9S	353.5E	–, –
MG-60	18°12.2′S, 47°05.4′E	2/2	232.4	54.5	–	–	40.8S	348.0E	–, –
MG-62 (LT)	18°27.2′S, 47°09.3′E	5/5	344.0	−21.1	20.1	17.5	72.8S	161.1E	9.7, 18.4
MG-62 (HT)		1/5	312.0	40.0	–	–	27.6N	356.5E	–, –
MG-64[a]	18°55.3′S, 47°42.6′E	12/13	80.0	−45.1	13.0	12.6	17.0S	340.6E	10.1, 15.9
MG-65[a]	18°35.3′S, 47°14.3′E	11/11	255.0	57.0	12	15	23.0S	351.0E	16.0, 21.0
MG-66 (LT)	18°38.4′S, 47°16.4′E	3/8	318.0	−26.0	121	11	49.5S	137.8E	6.4, 11.9
MG-66 (HT)[a]		6/8	314.0	32.0	14.0	19.0	32.0N	353.0E	12.0, 21.0
MG-67[a]	18°35.6′S, 47°12.1′E	3/3	132.0	−40.3	48.0	18.1	27.5N	356.6E	13.1, 21.8
MG-72 (LT)	18°43.9′S, 47°19.9′E	5/8	343.0	48.0	21.0	17.0	39.5N	28.0E	14.5, 22.2
MG-72 (HT)[a]		4/8	47.0	−32.0	25.0	19.0	45.4S	323.3E	12.0, 21.4
MG-74[a]	18°41.6′S, 47°17.3′E	7/8	293.0	56.0	42.0	9.0	06.0N	359.0E	10.0, 14.0
MG-84[a]	18°48.4′S, 47°24.9′E	4/4	258.5	39.4	9.0	31.0	17.7S	335.6E	22.2, 37.1
MG-86[a]	18°33.5′S, 47°12.6′E	3/5	286.0	56.0	42.0	20.0	01.0N	357.0E	21.0, 29.0
MG-90	18°11.6′S, 47°03.7′E	1/1	77.0	−38.0	–	–	18.2S	334.2E	–, –
MG-93 (LT)	17°56.8′S, 47°07.1′E	5/12	8.2	−16.9	69.0	9.6	77.7S	268.8E	5.1, 9.9
MG-93 (HT)		7/12	84.0	−30.0	9.0	21.0	10.4S	330.8E	12.9, 23.3
MG-113	−18°05.1′S, 47°10.7′E	8/9	292.0	46.1	23.0	12.0	10.0N	351.0E	9.8, 14.9
MG-114 (LT)	18°56.0′S, 47°32.1′E	7/11	343.0	−30.0	13.0	17.2	73.6S	144.8E	10.6, 19.1
MG-114 (HT)[a]		6/11	294.0	64.0	27.0	13.0	02.1N	007.1E	16.5, 20.7
MG-127 (LT)	19°09.8′S, 46°44.6′E	3/6	355.0	−20.0	136.0	10.6	79.9S	195.4E	5.8, 11.1
MG-127 (HT)[a]		2/6	85.0	−64.0	–	–	17.0S	001.0E	–, –
MG-130 (LT)	18°59.9′S, 47°16.5′E	7/7	324.0	−5.4	15.0	18.0	51.0S	157.4E	9.2, 18.3
MG-130 (HT)[a]		1/7	57.0	−32.0	–	–	36.1S	325.5E	–, –
MG-131[a]	19°01.6′S, 47°14.8′E	5/6	288.0	71.0	24.0	15.9	5.9S	14.4E	24.1, 27.7
MG-140 (LT)	18°52.2′S, 47°33.1′E	7/11	338.0	−22.0	14.0	16.7	67.0S	154.4E	9.3, 17.6
MG-140 (HT)[a]		3/11	77.0	−47.0	35.0	21.0	20.0S	341.0E	17.9, 27.0

n, number of samples used in calculating the mean; *N*, total number of samples from the site; kappa, precision parameter; a_{95}, cone of 95% confidence about the mean direction; VGP, virtual geomagnetic pole; (dp, dm), cone of 95% confidence about the paleomagnetic pole in the co-latitude direction (dp) and at a right angle to the co-latitude direction (dm). HT, high temperature/coercivity component; LT, low temperature/coercivity component.

[a] Sites are used in calculation of the mean pole within the D_2 thermal influence (virgation zone—see Fig. 2).

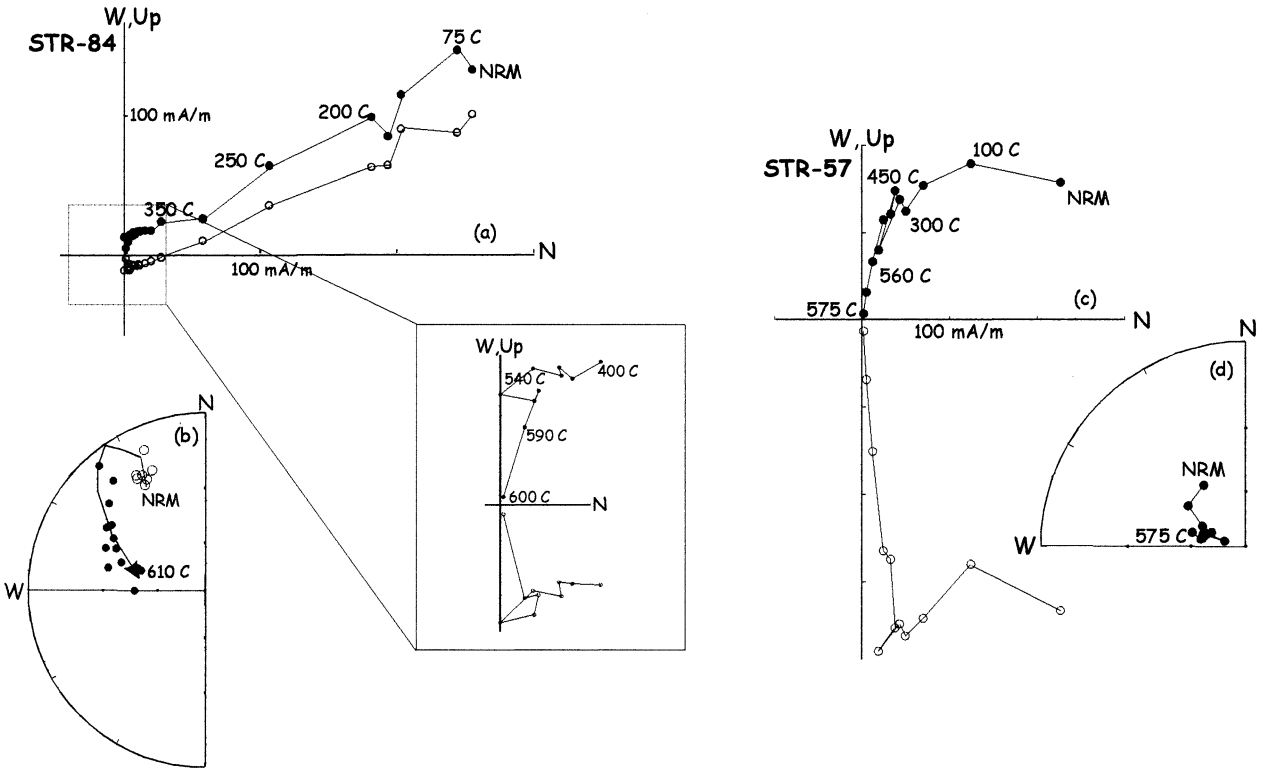


Fig. 4. (a–d) Zijderveld plots and associated equal-angle stereoplots of thermally treated samples STR-84 (site MG-114) and STR-57 (Site MG-131) showing the N–NE up overprint component and the high temperature characteristic component. Open (closed) circles represent the vertical (horizontal) vectors in the Zijderveld diagrams and open (closed) circles in stereonets represent up (down) inclinations.

($k = 16.3$, $a_{95} = 12.3°$; Fig. 6d and Table 2). Overall mean directions (after inverting the normal polarity directions) are Dec = 270.8°, Inc = 49.4° (samples, $k = 9$, $a_{95} = 5.1°$; Fig. 6e and Table 2) and Dec = 271°, Inc = 51.1° (sites, $k = 14.4$, $a_{95} = 10.1°$; Fig. 6f and Table 2). The McFadden and McElhinny (1990) reversal test for sites shows that the angle between the normal and reverse directions is 11.7° and the reversal test is indeterminate. Virtual geomagnetic poles were calculated for each site and the mean of the virtual geomagnetic poles (VGPs) falls at 8.9°S, 348.3°E ($a_{95} = 11.8°$; Tables 1 and 2).

In addition to the sites outlined above, we sampled several of the Ambatomiranty dykes, but only one site yielded a stable magnetization (MG-132). Fig. 7a and b show typical demagnetisation curves for the Ambatomiranty samples from that site. A VGP calculated from the mean direction (Dec = 286°, Inc = +45°, $k = 521$, $a_{95} = 3.4°$; Fig. 7c) falls at 5° N, 347°E (dp = 2.7°, dm = 4.3°).

The characteristic high temperature directions observed in the stratoid granites unblock between 400 and 600 °C, with the majority of the vectors isolated at temperatures between 500 and 580 °C (Figs. 4, 5 and 8a). Alternating field demagnetisation resulted in a rapid decrease in intensity for most samples (median destructive field between 10 and 40 mT), followed by a relatively hard coercivity component at higher fields (Fig. 8b). These demagnetisation characteristics along with isothermal remanence acquisition studies that show saturation between 0.1 and 0.4 T, suggest that the main carrier of the characteristic remanence is low Ti-magnetite (Fig. 8c).

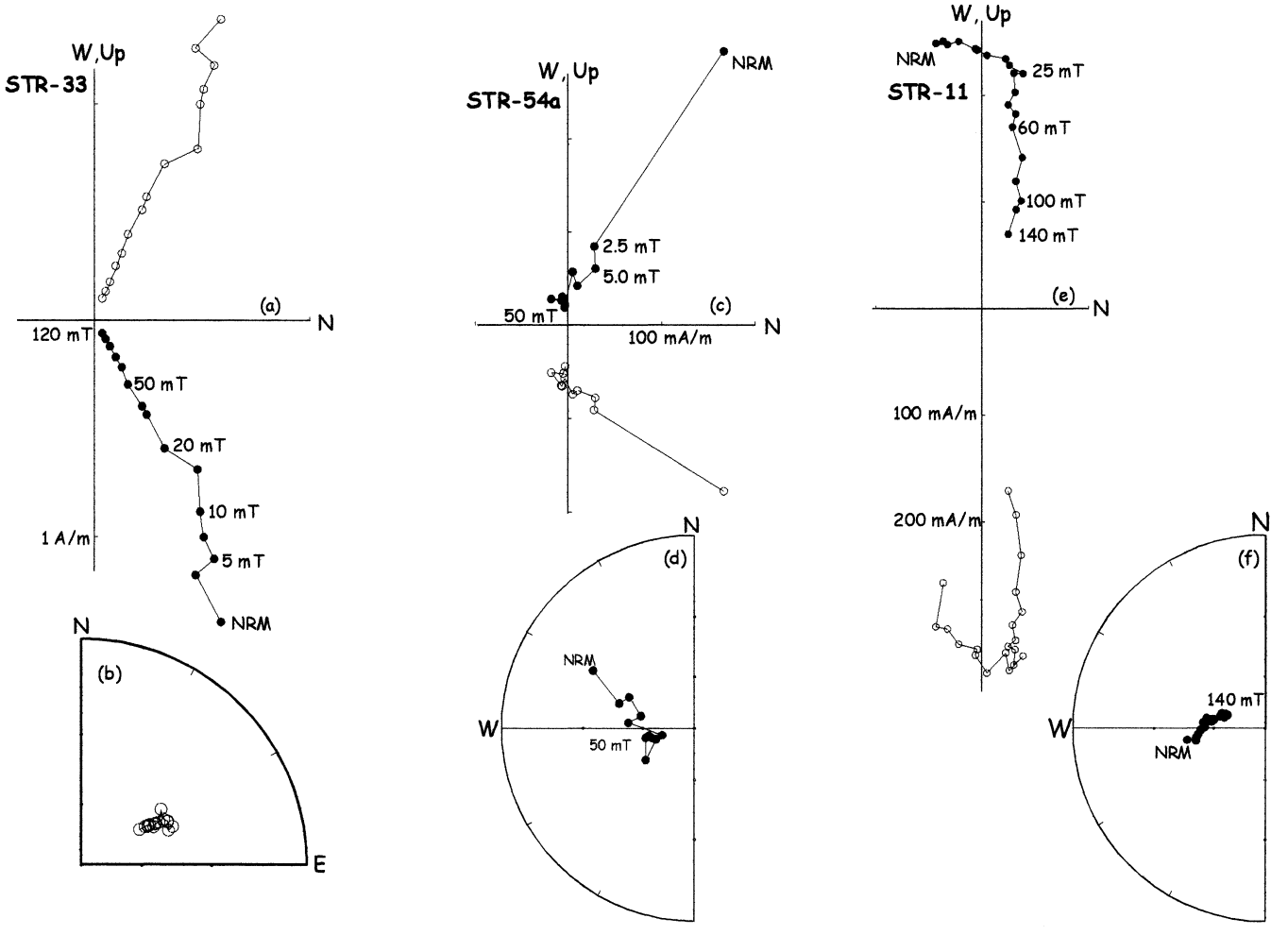


Fig. 5. (a–f) Zijderveld plots and associated equal-angle stereoplots of alternating field treated samples STR-33 (site MG-65) and STR-54a (Site MG-131) and STR-11 (Site MG-113) showing both normal (STR-33) and reverse (STR-54a, STR-11) polarities of the characteristic component. Open (closed) circles represent the vertical (horizontal) vectors in the Zijderveld diagrams and open (closed) circles in stereonets represent up (down) inclinations.

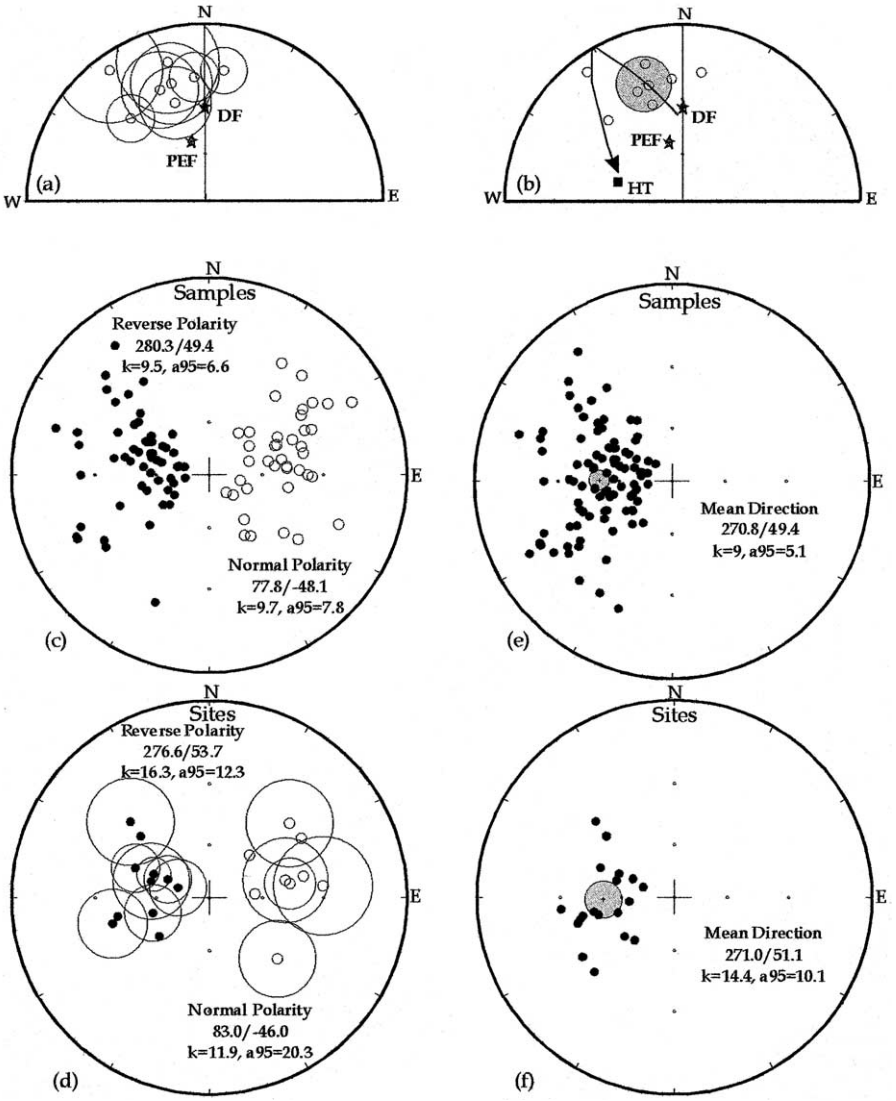


Fig. 6. (a) Site mean overprint directions, DF, Dipole field expected direction and PEF, present Earth's field direction based on the 1995 IGRF. (b) Mean overprint direction showing the great circle relationship between the DF and the HT (high-temperature/coercivity direction). (c) Individual sample directions based on high temperature/coercivity components as both normal and reverse polarity. (d) Site mean directions and a_{95} circles for sites ($N \geqslant 2$ samples) showing both normal and reverse polarity. (e) Sample mean direction after inverting the normal polarity directions and (f) site mean direction after inverting the normal polarity directions. Open (closed) circles in stereonets represent up (down) inclinations.

3.2. $^{40}Ar/^{39}Ar$ geochronology

Although U–Pb ages can be reasonably correlated to the timing of primary remanence acquisition in rapidly cooled igneous rocks, slowly-cooled plutonic rocks generally acquire partial thermal remanences over a wide range of temperatures during the cooling of the pluton. As we argue below, the original emplacement of the stratoid granites north of Antananarivo took place at about 630 Ma (Paquette and Nédélec, 1998), but some of the rocks were subsequently reheated at

Table 2
Mean results stratoid granites

Details	Declination	Inclination	k	a_{95}	Pole latitude	Pole Longitude	dp	dm
Overprint component								
Sites—secondary component (9 sites)	342.0	−19.8	23.1	11.8	70.7°S	160.3°E	6.5	12.3
Main component—samples								
(1) Samples—reverse polarity	280.3	+49.4	9.5	6.6	0.6°S	348.8°E	5.8	8.7
(2) Samples—normal polarity	77.8	−48.1	9.7	7.8	19.2°S	342.6°E	6.7	10.2
(3) Samples—mean all ($N=94$)	270.8	+49.4	9.0	5.1	8.5°S	346.4°E	4.5	6.8
Main component—sites ($N \geqslant 2$ samples)								
Sites—reverse polarity ($N=10$)	276.6	+53.9	16.3	12.3	5.0°S	352.0°E	12.1	17.2
Sites—normal polarity ($N=6$)	83.0	−46.0	11.9	20.3	14.4°S	341.8°E	16.6	26.0
(4) Sites—mean all ($N=16$)	271.0	+51.1	14.4	10.1	8.8°S	348.0°E	9.3	13.7
Sites north of virgation ($N \geqslant 2$ samples)								
Mean VGPs ($N=4$)				27.2	15.4°S	340.1°E		
Sites in virgation zone ($N \geqslant 2$ samples)								
(5) Mean VGPs ($N=12$)				14.2	6.7°S	352.6°E		
Main component—sites ($N \geqslant 2$ samples)								
(6) Sites-mean of VGPs				11.8	8.9°S	349.0°E		
Ambatomiranty dike (5 samples)	286.0	+45.0	521	3.4	5.0°N	347.0°	2.7	4.3
Main component (Africa reference)								
Samples mean				5.1	9.7°N	345.1°E		
Sites mean				9.5	9.6°N	346.8°E		
Mean VGPs				11.8	9.7°N	348.3°E		
Mean virgation sites				14.2	12.2°N	351.1°E		
Ambatomiranty dike				3.4	23.2°N	347.0°E		

Madagascar is rotated to East Africa using an euler pole at: 3.41°S, 81.70°W, +19.73° (ccw). Poles are calculated based on a common site at 18.5°S, 47.3°E except for the VGP mean. Numbers refer to poles shown in Fig. 13.

about 550 Ma. This reheating reset the argon systematics and therefore the cooling trend observed in the U–Pb and $^{40}Ar/^{39}Ar$ data reflects either a cooling stage after the D_1 event and magma emplacement (at 630 Ma), or the reheating and second cooling stage as a consequence of the D_2 event.

The minerals were taken from the same samples used for paleomagnetic studies through standard mineral separation techniques. Specifically, we removed hornblende from sample STR-57 (MG-131); biotites from samples STR-11 (MG-113), STR-54 (MG-131), STR-33 (MG-65) and STR-84 (MG-114) along with potassium feldspar from sample STR-84. The samples were then packaged and sent to the irradiation facilities at the University of Michigan. The irradiation standard used was hornblende MMHb-1 with an assumed K–Ar age of 520.4 Ma (Samson and Alexander, 1987). Each irradiation consisted of a set of samples and standards sealed within evacuated fused silica tubes, and from each tube, at least 3 analyses for each of 5 packets of MMHb-1 were analysed. The resulting values of J were interpolated for each unknown sample location within the reactor. Error estimates for J include scatter about the interpolation function. Individual run data are included in Appendix A.

All argon age analyses were performed using a VG1200S mass spectrometer equipped with a Daly detector operated in analog mode. Samples were step-heated using a coherent INNOVA model 70 argon ion laser with a nominal maximum output power of 5 W. All analyses were performed with a fully automated fusion system, with laser power being set under computer control. Step-heating durations were 60 s, and the total fusion plus gas clean-up time was 3 min. Laser fusion system

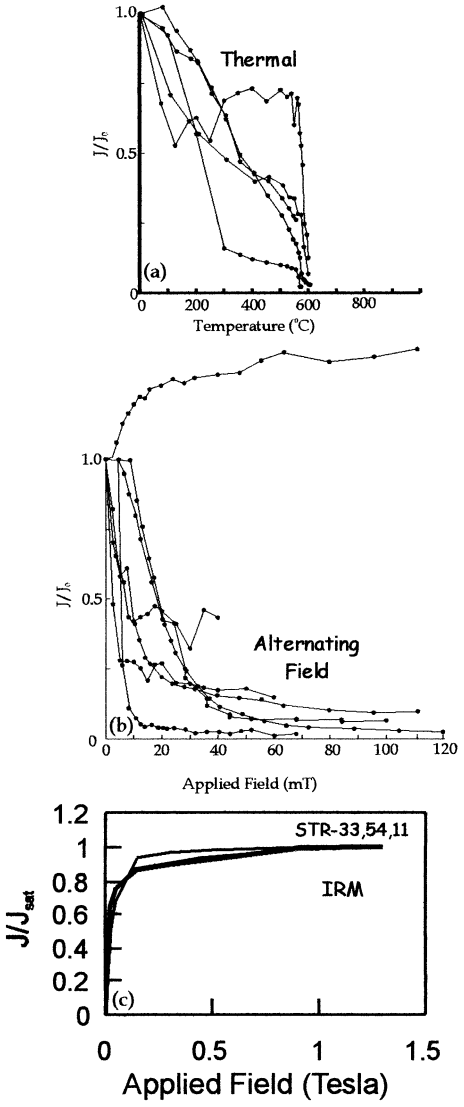


Fig. 7. (a) Thermal demagnetisation behavior of selected samples from the stratoid granites showing hard-shoulder behavior at temperatures above 500 °C. Samples showing the overprint direction generally show more distributed J/J_0 (normalized intensity) plots at lower temperatures. (b) Alternating field demagnetisation behavior of selected samples showing the relatively low median destructive field but fairly hard coercivity component at higher fields (see also Fig. 5). (c) Isothermal remanence acquisition studies of AF-treated samples showing saturation by 0.4 T characteristic of magnetite.

blanks were monitored frequently (typically every 5–6 fractions) and were normally about 1×10^{-13}, 3×10^{-13}, 3×10^{-14}, 3×10^{-14}, and 5×10^{-12} mISTP for masses 36 through 40 respectively. Mass discrimination in the mass spectrometer was monitored daily by analysing an aliquot of atmospheric argon with a volume of about 5×10^{-9} mISTP. Plateau ages are calculated as an error weighted mean where MSWD < 2 and over 50% of the ^{39}Ar was represented in the overlapping age spectral segment. In cases where plateau criteria were not met, an integrated age is reported which represents a weighted mean age summed over the nearly flat portion of the spectrum. All errors are 1σ unless otherwise noted. The high radiogenic yields of the samples preclude the generation of statistically valid isochron plots and therefore integrated or plateau ages from the samples are preferred.

Fig. 9a, d and g show stepwise degassing results from biotites removed from samples STR-33 (Site MG-65), STR-54 (Site MG-131) and STR-84 (Site MG-114). Each of these samples was divided into two splits and both degassing runs are shown in the figures (results listed in Table 3). Sample STR-33 (Fig. 9a) yielded plateau ages of 502.4 ± 0.65 Ma (88% gas release; MSWD = 2.56) and 503.5 ± 0.65 Ma (82% gas release; MSWD = 10.3). An inverse isochron analysis on the splits yielded ages of 502.4 ± 0.72 and 504.2 ± 0.85 Ma (both with low ^{40}Ar/^{36}Ar initial ratios, Fig. 9b, c; Table 3). Sample STR-54 (Fig. 9d) yielded very well-defined plateau ages of 499.8 ± 0.6 Ma (97% gas release; MSWD = 0.96) and 499.7 ± 0.6 Ma (94% gas release; MSWD = 1.19) with nearly identical inverse isochron ages and ^{40}Ar/^{36}Ar initial ratios near the atmospheric ratio (Fig. 9e and f; Table 3). Sample STR-84 (Fig. 9g) showed a small amount of argon loss at lower laser powers and neither split showed a well-defined plateau. Integrated ages over the final 66–75% of the gas yielded ages of 494.5 ± 0.9 (MSWD = 3.38) and 496.0 ± 0.51 Ma (MSWD = 1.9), respectively. Inverse isochrons showed considerable scatter of data points and gave ages of 494.9 ± 1.7 Ma (^{40}Ar/^{36}Ar initial ratio of 245.5) and 494.7 ± 2.5 Ma (^{40}Ar/^{36}Ar initial ratio of 543.6; Fig. 9h and i; Table 3). Collectively, a mean biotite age for the stratoid granites in the virgation zone is calculated as the mean of the plateau and integrated ages of the individual splits at 499.3 ± 3.5 Ma (Table 3).

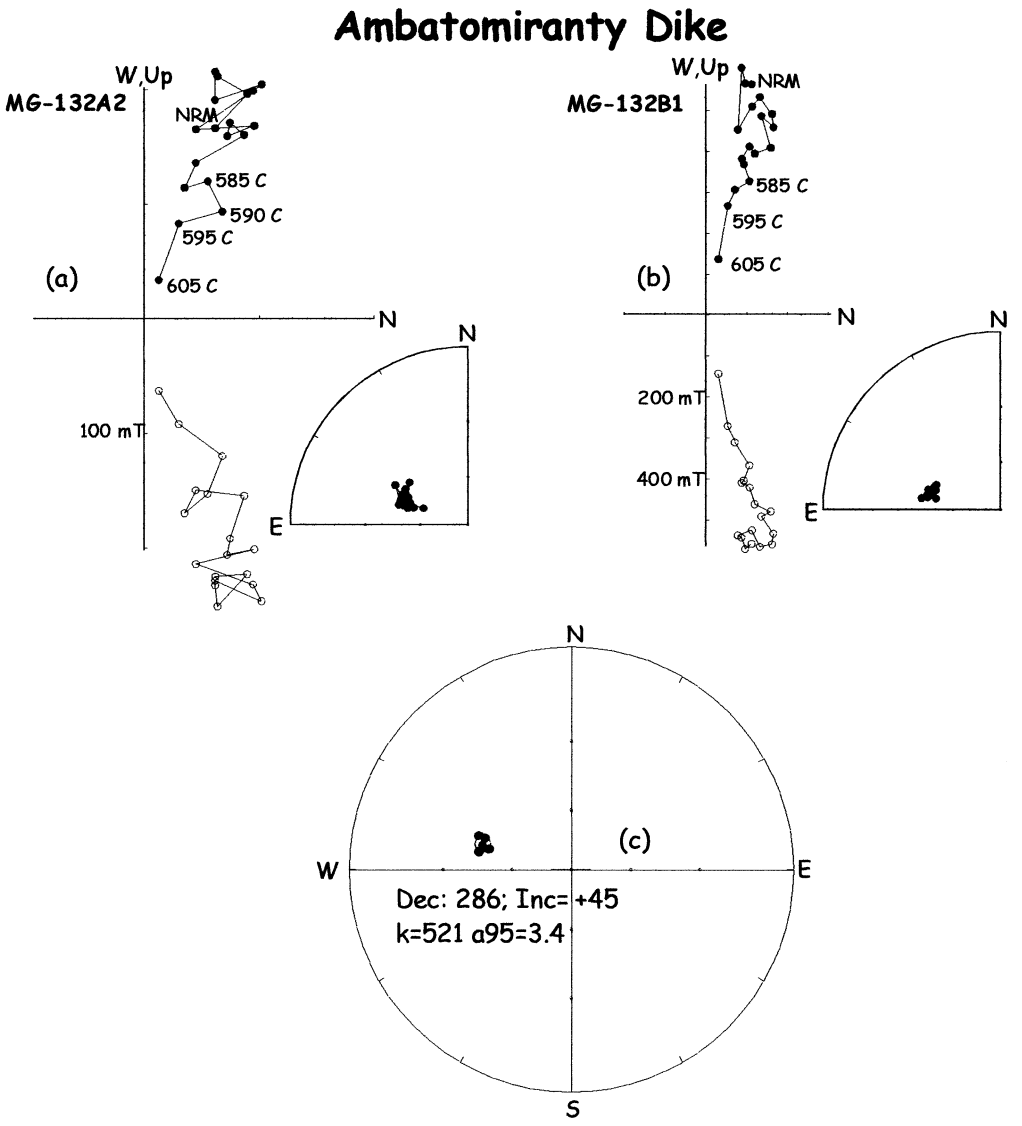


Fig. 8. (a–b) Zijderveld plots and associated equal-angle stereoplots of thermally treated samples from the Ambatomiranty dykes (Site MG-132). Open (closed) circles represent the vertical (horizontal) vectors in the Zijderveld diagrams and open (closed) circles in stereonets represent up (down) inclinations. (c) Mean direction from the Ambatomiranty dyke sampled at Site MG-132.

Fig. 10a shows stepwise degassing results from the sample of hornblende (STR-57, site MG-131; Table 3). A small amount of excess argon is observed at low laser power, but the sample showed a well-defined plateau of 515.7 ± 0.94 Ma (85% gas release; MSWD = 0.30). The inverse isochron age for this sample is 514.8 ± 3.2 Ma with $^{40}Ar/^{36}Ar$ initial ratio of 399.1 (Fig. 10c; Table 3).

Biotite grains removed from sample STR-11 (Site MG-113) yielded much older ages than the samples described above (Fig. 10c; Table 3). Although each split showed argon loss at low temperatures, both yielded high temperature inte-

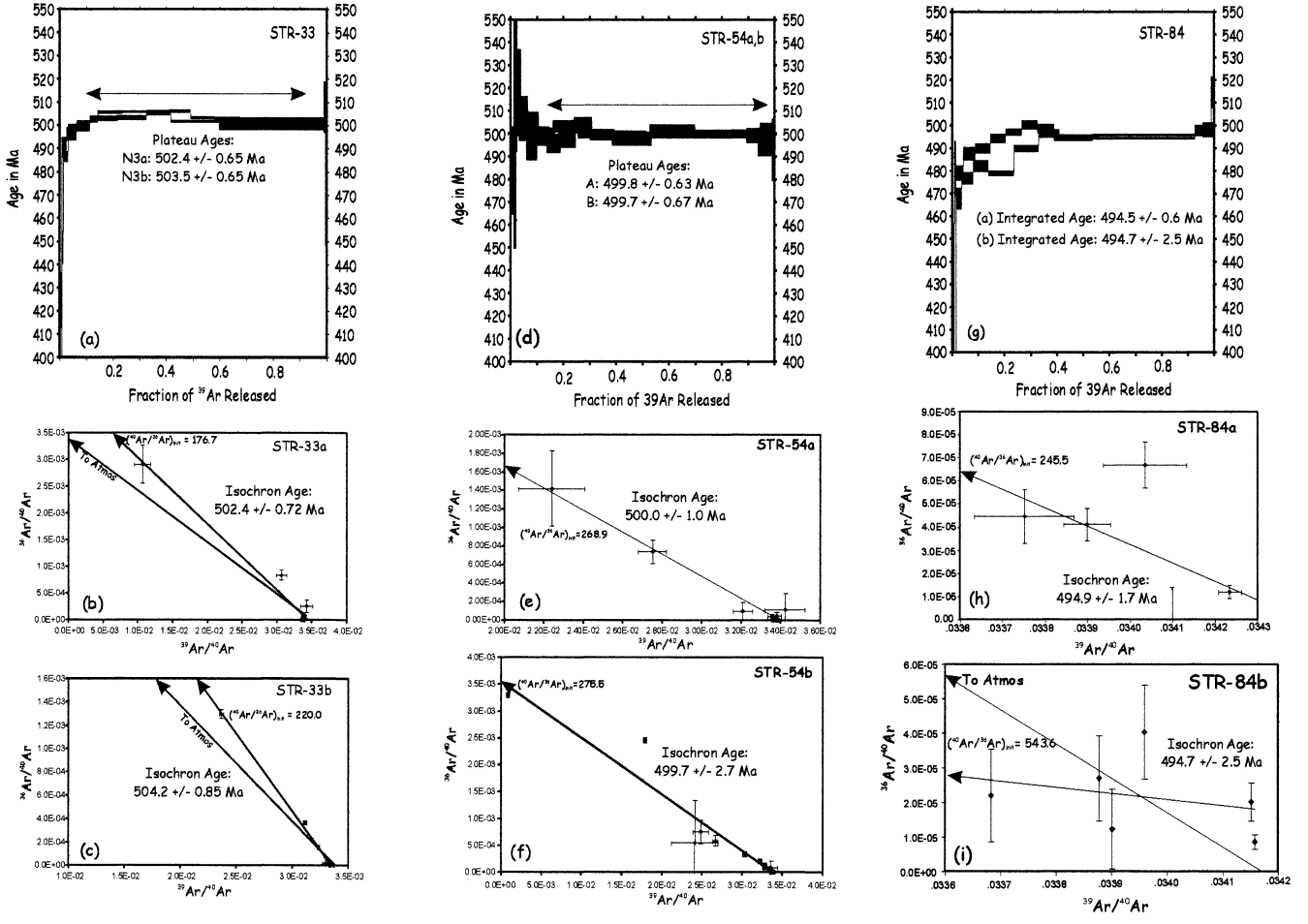


Fig. 9. (a) Stepwise degassing spectra for biotite sample STR-33 (2 splits; Site MG-65) along with (b, c) inverse isochron diagrams (atmos = atmospheric initial $^{36}Ar/^{40}Ar$ ratio of 295.5). (d) Stepwise degassing spectra for biotite sample STR-54 (2 splits; Site MG-131) along with (e and f) inverse isochron diagrams for both splits. (g) Stepwise degassing spectra for biotite sample STR-84 (2 splits; Site MG-114) along with (h, i) inverse isochron diagrams for both splits.

Table 3
$^{40}Ar/^{39}Ar$ results

Sample name/site	J-value ± error	Total gas age ± error	Plateau (P) or integrated age (I) ± error	MSWD	Isochron age ± error	$(^{40}Ar/^{36}Ar)_{init}$
STR-33a/MG-65 (bt)	$1.08358e^{-2} \pm 1.575e^{-5}$	500.1 ± 1.28 Ma	502.4 ± 0.65 Ma (P)	2.56	502.4 ± 0.72 Ma	176.7
STR-33b/MG-65 (bt)	$1.08358e^{-2} \pm 1.575e^{-5}$	500.7 ± 0.68 Ma	503.5 ± 0.65 Ma (P)	10.3	504.2 ± 0.85 Ma	220.0
STR-54a/MG-131 (bt)	$1.08221e^{-2} \pm 1.624e^{-5}$	498.9 ± 1.07 Ma	499.8 ± 0.63 Ma (P)	0.96	500.0 ± 1.00 Ma	268.9
STR-54b/MG-131 (bt)	$1.08221e^{-2} \pm 1.624e^{-5}$	497.2 ± 1.01 Ma	499.7 ± 0.67 Ma (P)	1.19	499.7 ± 2.70 Ma	275.5
STR-84a/MG-114 (bt)	$1.08358e^{-2} \pm 1.787e^{-5}$	493.6 ± 0.89 Ma	494.5 ± 0.60 Ma (I)	3.38	494.9 ± 1.70 Ma	245.5
STR-84b/MG-114 (bt)	$1.08358e^{-2} \pm 1.787e^{-5}$	488.5 ± 0.86 Ma	496.0 ± 0.51 Ma (I)	1.90	494.7 ± 2.50 Ma	543.6
STR-57a/MG-131 (hbl)	$1.07991e^{-2} \pm 1.414e^{-5}$	530.5 ± 2.70 Ma	515.7 ± 0.96 Ma (P)	0.294	514.8 ± 3.20 Ma	399.1
STR-84c/MG-114 (K-spar)	$1.08519e^{-2} \pm 2.003e^{-5}$	445.0 ± 0.77 Ma	–		–	–
STR-11a/MG-113 (bt)	$0.94201e^{-2} \pm 2.969e^{-5}$	609.0 ± 1.66 Ma	611.9 ± 1.7 Ma (I)	1.42	612.0 ± 1.70 Ma	292.9
STR-11b/MG-113 (bt)	$0.94201e^{-2} \pm 2.969e^{-5}$	598.1 ± 1.62 Ma	607.9 ± 1.7 Ma (I)	1.27	607.3 ± 1.70 Ma	427.3

bt, biotite; hbl, hornblende; K-spar, potassium feldspar; MSWD, mean square of weighted deviates.

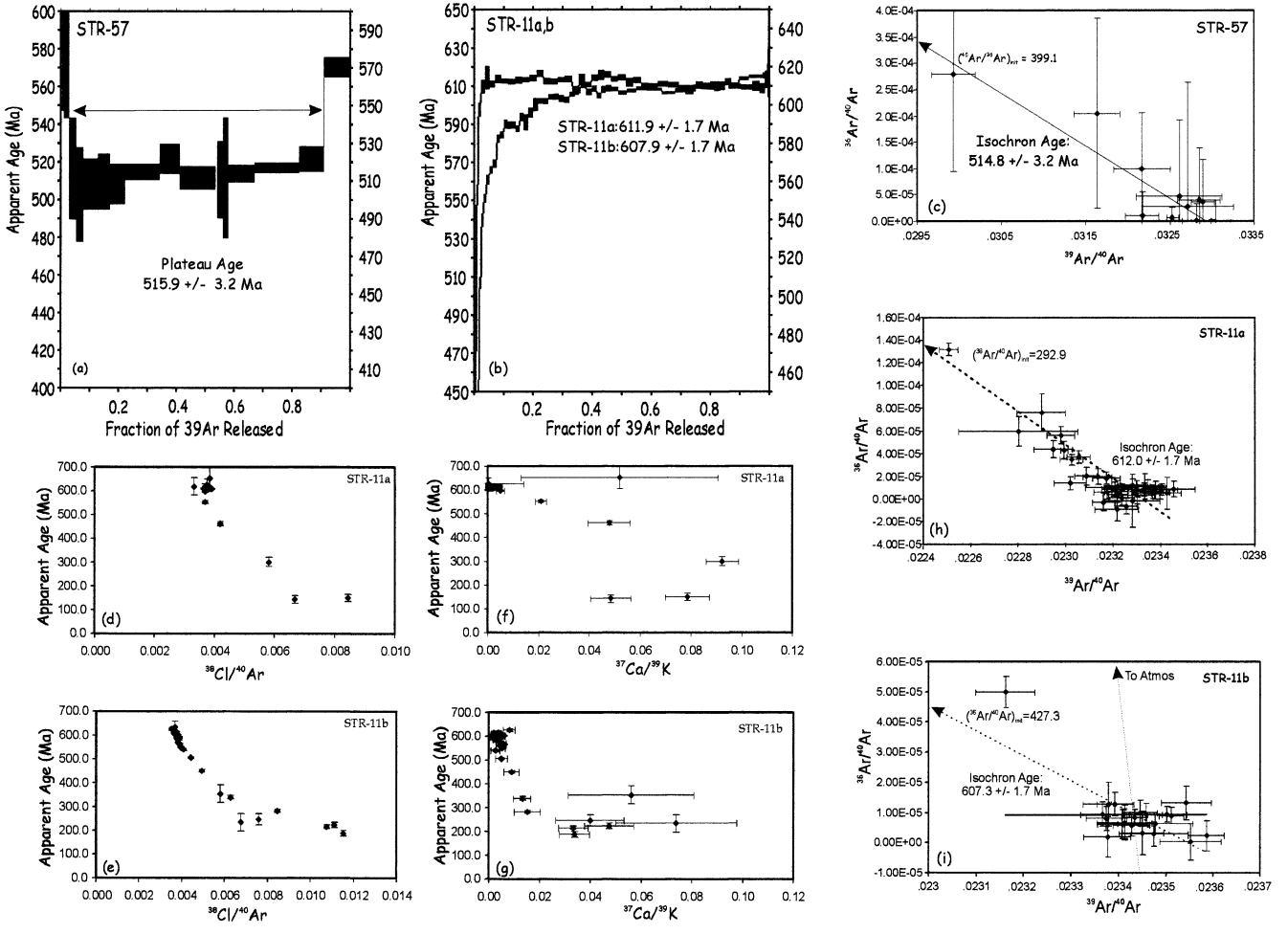


Fig. 10. (a) Stepwise degassing spectra for hornblende sample STR-57 (1 splits; Site MG-131). (b) Stepwise degassing spectra for biotite sample STR-11 (2 splits; Site MG-113) (c) inverse isochron diagram for sample STR-57 (atmos = atmospheric initial $^{36}Ar/^{40}Ar$ ratio of 295.5). (d) Plot of $^{38}Cl/^{40}Ar$ for sample STR-11a showing alteration associated with lower apparent ages (note: errors on the $^{38}Cl/^{40}Ar$ ratios less than 0.006 are larger than the reported ratio, but the trends suggest minor chlorite alteration of the biotites that is removed at low laser power) (e) plot of $^{37}Ca/^{39}K$ for STR-11a also showing a correlation between higher $^{37}Ca/^{39}K$ ratios and mostly lower apparent ages (f) plot of $^{38}Cl/^{40}Ar$ for sample STR-11b showing alteration associated with lower apparent ages and (g) plot of $^{37}Ca/^{39}K$ for STR-11a also showing a correlation between higher $^{37}Ca/^{39}K$ ratios and mostly lower apparent ages (h and i) inverse isochron diagrams for both splits of STR-11.

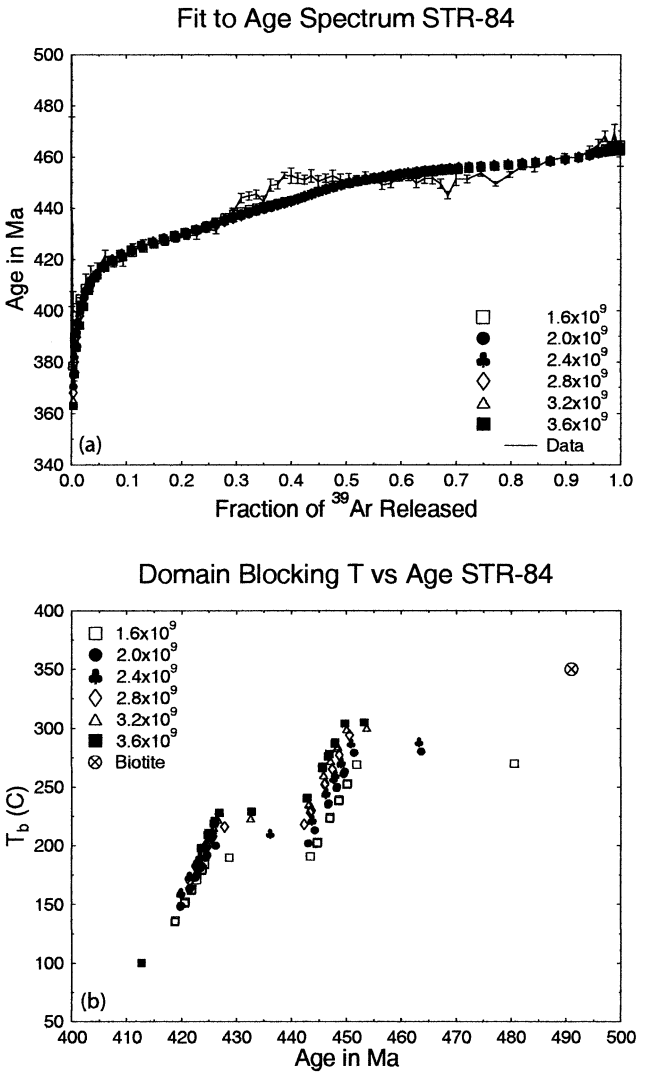


Fig. 11. (a) Results from analyses of the K-feldspar sample STR-84 showing the best fit multi-domain age spectral models for the run. Only data with laser powers <2000 mW and less were fit, and the fractions above that level were combined into a single fraction. There is very little difference between models with different values of α. (b) Plot of the multi-domain model domain closure temperatures (assuming slab geometry and cooling rate at blocking of 3 °C/Ma) for STR-84 K-feldspar. A range of α values for each run are shown. For comparison, the integrated gas age for boitite STR-84 is shown with an assumed closure temperature of 350 °C.

grated ages. The first split yielded an integrated age of 611.9 ± 1.7 Ma over 96% of the gas (MSWD = 1.42). The second split yielded a slightly lower integrated age of 607.9 ± 1.7 Ma over 77% of the gas (MSWD = 1.27; Fig. 10b). Ratios of $^{37}Ca/^{39}K$ and $^{38}Cl/^{40}Ar$ indicate minor alteration of the biotite resulting in lower apparent ages at low laser powers (Fig. 10d–g). The inverse isochron plots showed that the second sample had a slightly higher $^{40}Ar/^{36}Ar$ initial ratio of 427.3 (versus 292.9) and significantly more argon loss at low laser power (Fig. 10h, i; Table 3). Neither sample showed unequivocal evidence for excess argon although the second split showed a high

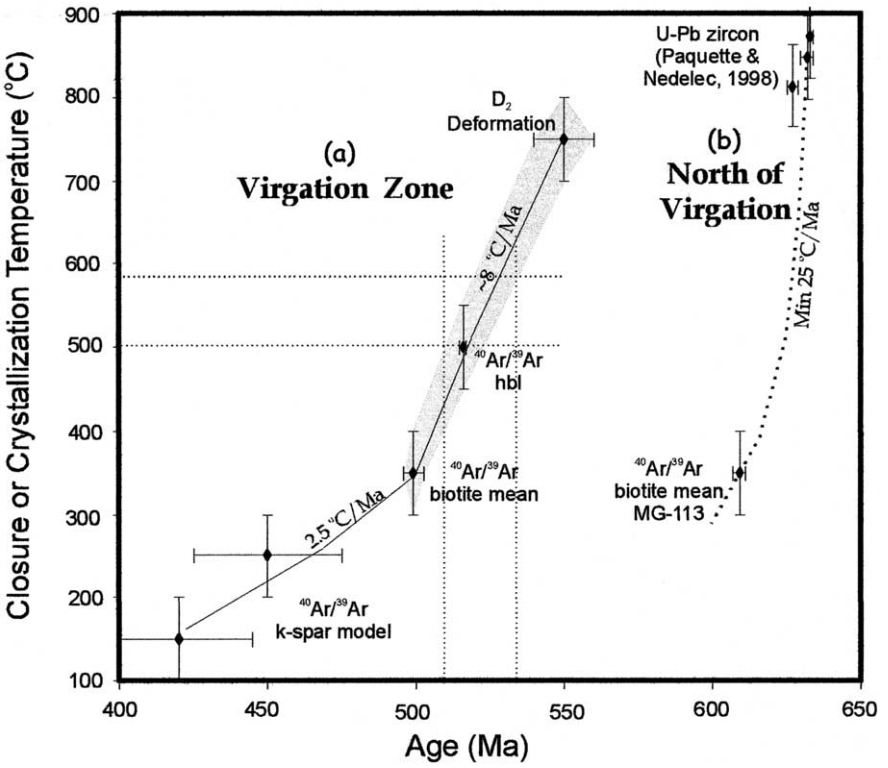


Fig. 12. (a) Cooling curve for virgation zone stratoid granites based on the $^{40}Ar/^{39}Ar$ values reported here and a deformation age of ~550 Ma for samples within the virgation zone (see text and Figs. 2 and 3 for locations of samples). The initial cooling rate is ~8 °C/Ma and slows to about 2.5 °C/Ma. (b) Hypothetical cooling curve for stratoid granites north of the virgation zone based on Paquette and Nédélec (1998) from sites MG-65, MG-14 and MG-93 (Fig. 2) and the text for complete explanation. An estimated cooling rate of 25 °C/Ma is implied by these data.

$^{40}Ar/^{36}Ar$ initial ratio. These data suggest a biotite age of ~610 Ma for the stratoid granite at Site MG-113.

Potassium feldspar (K-spar) grains were also separated from sample STR-84. Meert et al. (2001b) describe a method for evaluating the domain blocking temperatures in laser-heated K-spar samples by first calculating a range for the proportionality constant α based on laser power and the Stefan–Boltzman law for black-body radiation. The maximum blocking temperature of 350 °C for the K-spar was constrained by the fact that the maximum K-feldspar age is lower than the co-existing biotite ages, and 350 °C is a reasonable blocking temperature estimate for biotite. Fig. 11a shows the modelled release pattern compared with the actual K-spar data and Fig. 11b shows the modelled domain blocking temperatures for our sample. The data indicate cooling from 350 °C at ~495 Ma (age of the STR-84 biotite) to 250 °C at ~450 Ma (2.2 °C/Ma) and 150 °C at ~420 Ma (3.3 °C/Ma) assuming a simple exhumation path.

4. Discussion

4.1. Age of paleomagnetic pole

The paleomagnetic and $^{40}Ar/^{39}Ar$ data from the stratoid granites suggests a pervasive remagnetization event in central Madagascar coeval with late shearing in the Angavo Belt and ductile deformation in the Antananarivo virgation zone. The maximum age for this remagnetization is constrained by a number of U–Pb and Pb–Pb ages (Paquette and Nédélec, 1998; Tucker et al., 1999a;

Kröner et al., 1999, 2000) in the area between 550 and 561 Ma (D_2 and D_3 events of Nédélec et al., 2000).

The $^{40}Ar/^{39}Ar$ data obtained in this study allow us to evaluate the thermal conditions during remagnetization of the stratoid granites. Fig. 12 shows possible cooling trends in the region based on data reported in this paper and the papers by Kröner et al. (2000, 1999), Paquette and Nédélec (1998). Two trends are noted, the first (younger trend) shows results from samples collected in the virgation zone and the second (older trend) is for rocks collected in the northern D_1 domain. Both trends assume a simple cooling pattern following either intrusion or metamorphism. For the magmatic cooling trend, we assigned magmatic temperatures for the samples MG-93, MG-14 and MG-65 of 872, 847 and 813 °C, respectively (calculated after Watson and Harrison, 1983). We note that sample MG-113, from which the biotite age is produced, belongs to the same petrogenetic source as MG-65 (Nédélec et al., 1995). Therefore, the cooling trend depicted here is not strictly for a single stratoid granite body, but represents an idealized curve for the northern stratoid granites. For the metamorphic trend, incipient migmatization and upper amphibolite facies metamorphism suggest maximum temperatures of around 750 °C (Nédélec et al., 2000). A closure temperature of 500 ± 50 °C was assigned to the homblende in sample STR-54 (site MG-131). The characteristic magnetization recognized in this study was blocked at temperatures between 500 and 580 °C. If the magnetization in the stratoid granites resulted solely from slow-cooling following high-temperature D_2 event in the Antananarivo virgation zone, then the age of the magnetization is constrained by this cooling curve at 521.4 ± 11.9 Ma.

We expected that paleomagnetic samples characterized by the D_1 structural pattern would yield an older magnetization. This was expected, for example, at site MG-113 based on the ~610 Ma biotite age described above. An exponential fit to the cooling curve in Fig. 12 based on our biotite data and the zircon ages reported in Paquette and Nédélec (1998) yields an approximate cooling rate of 25 °C/Ma for the northern stratoid granites between 630 and 610 Ma assuming a simple exhumation path following intrusion. Unfortunately, even paleomagnetic samples whose structures were not rotated by D_2 might have been reset by this thermo-chemical event. For example, at site MG-65 (the nearest sample to the D_2 virgation zone) both the $^{40}Ar/^{39}Ar$ biotite ages and the paleomagnetic directions appear to be reset by D_2 although the rocks show no structural rotation consistent with D_2 deformation. Apparently, the thermal influence of the D_2 event diminishes between site MG-65 and MG-113 but still had a profound effect on the paleomagnetic directions.

Alternatively, the $^{40}Ar/^{39}Ar$ biotite ages may contain undetectable excess argon and therefore the actual ages may be younger (Gaber et al., 1988; Reddy et al., 1996). While we cannot rule out such a scenario, we note that whole-rock ^{87}Rb–^{86}Sr isochron ages from chemically similar stratoid granites in the region yield an age of 643 ± 17 Ma (MSWD = 2.1) and stratoid granites from the strongly alkaline suite yielded an isochron age of 623 ± 14 Ma (MSWD = 1.2; Nédélec et al., 1999). Thus, we view excess argon as a less-likely explanation for the age patterns in biotites from MG-113.

Fig. 13a shows the mean poles (keyed to the numbers listed in Table 2) and Fig. 13b shows the mean poles rotated to African coordinates (following Meert et al., 2001a). In this case, a mean of VGPs south of site MG-65—where either/both structural or geochronologic evidence suggests imprinting of the D_2 event—yields a pole at 6.7°S, 352.6°E ($a_{95} = 14.2°$). The approximate age of this pole is 521.4 ± 11.9 Ma.

4.2. Tectonic implications

Our new $^{40}Ar/^{39}Ar$ ages, combined with a number of recent U–Pb and Pb–Pb ages and structural data, provide important constraints on the tectonic history of Madagascar and, by inference, Gondwana assembly. The ~610 Ma $^{40}Ar/^{39}Ar$ age for biotites taken from a stratoid granite (characterized by unrotated D_1 structural pattern) north of the virgation zone suggests that part of Madagascar escaped the thermal imprint of the D_2–D_3 events recorded further south. This

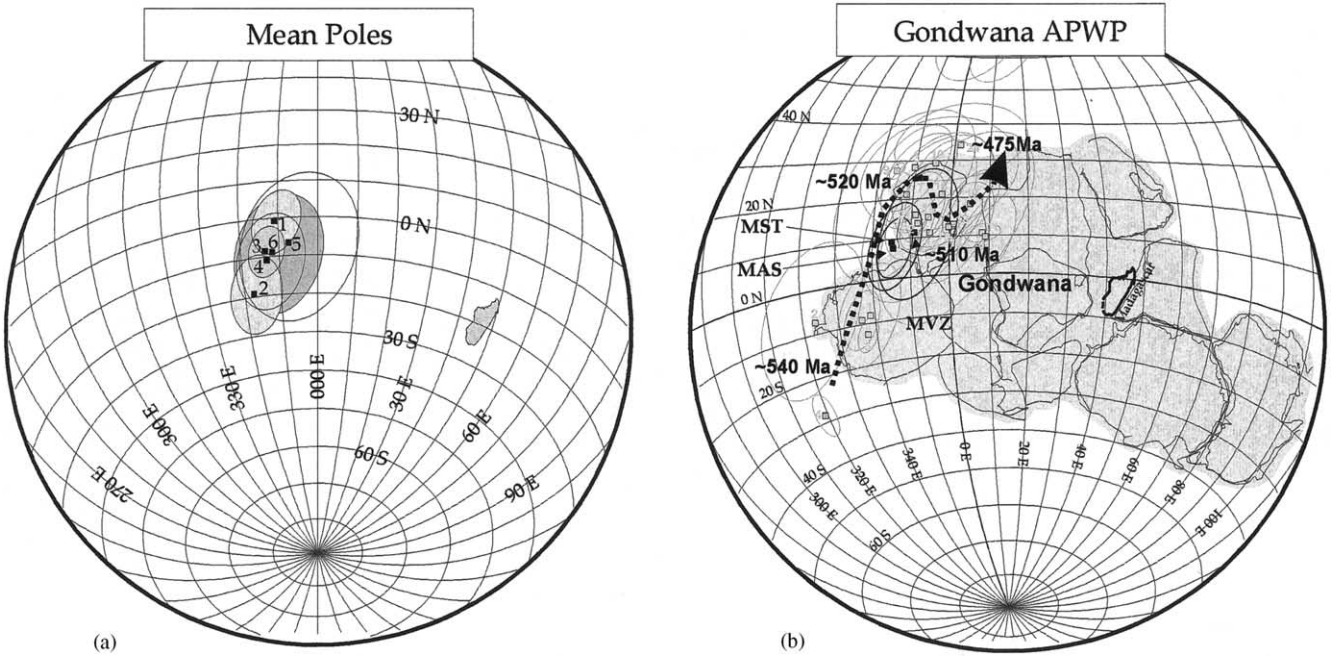


Fig. 13. (a) Mean poles from this study in an African reference frame according to the rotation parameters of Lawver and Scotese (1987): (1) Mean pole calculated from all reverse plarity samples; (2) Mean pole based on all normal polarity samples; (3) Mean pole calculated from all sample directions; (4) Site mean pole calculated from sites with $N \geqslant 2$ samples; (5) Mean VGP for sites within the virgation zone; and (6) VGP mean of all sites in this study with $N \geqslant 2$ samples. (b) Mean poles rotated to Gondwana coordinates as follows; MAS, mean of all samples; MST, mean of all sites ($N \geqslant 2$ samples) and MVZ, mean virgation zone pole (estimated age $521.4.0 \pm 11.9$ Ma).

interpretation is supported by preliminary results from Rb–Sr whole rock isochrons on the northern stratoid granites, but remains controversial in light of numerous ~550 Ma ages throughout central Madagascar (Kröner et al., 2000).

Paquette and Nédélec (1998) built on the earlier work by Nédélec et al. (1994, 1995) and suggested that the stratoid granites of central Madagascar were emplaced during a phase of extensional tectonism following the collision of Madagascar with eastern Africa. They estimated this collision occurred between 700 and 650 Ma and that the post-collisional extension occurred around 630 Ma. The rapid cooling registered by the northern stratoid granites ($\geqslant 25$ °C/Ma) is consistent with a high transient geothermal gradient due to advection of magmatic heat in the middle crust, followed by rapid unroofing of the basement. The timing of this extension is broadly coeval with post-collisional extension observed in the East African Orogen (Stern, 1994; Cosca et al., 1999; Meert, in press) although it is clearly older than the extensional features noted elsewhere in Madagascar (De Wit et al., 2001; Collins et al., 2000).

There is considerable debate about the timing of Gondwana assembly in eastern Africa and East Gondwana (Meert, in press, De Wit et al., 2001; Kröner et al., 2000; Möller et al., 2000; Muhongo et al., 2001; Paquette and Nédélec, 1998; Appel et al., 1998; Meert and Van der Voo, 1997; Kriegsman, 1995; Stern, 1994; Collins and Windley, 2002). One suggestion, put forth originally by Meert et al. (1995) is that East Gondwana is simply a geographically convenient term for an amalgam of cratonic elements whose Neoproterozoic suturing occurred during two main orogenic pulses. The East African Orogeny (Stern, 1994) reflects not only the suturing of arc terranes in the Arabian–Nubian shield, but also collision between Madagascar, India, Seychelles, Sri Lanka, India and coastal regions of East Antarctica with eastern Africa culminating between 620 and 650 Ma. This was followed some 100–120 million years later by the collision of Antarctica and Australia with the previously assembled blocks during an event termed the Kuunga Orogeny. Fitzsimons (2000) recently identified a region of major orogenesis in East Antarctica during the 500–570 Ma interval that was ascribed to both (a) final amalgamation of East Antarctica (e.g. Mawson continent and coastal regions of East Antarctica) and (b) its collision with eastern Africa. Our new Ar–Ar data, as well as the recent geochronologic investigations by De Wit et al. (2001) in southern Madagascar along with the work of Muhongo et al. (2001) in Tanzania, support previous assertions by Paquette and Nédélec (1998) for a ~650 Ma onset of collision. However, that conclusion contrasts sharply with the interpretations of Appel et al. (1998), Möller et al. (2000) who recognize the existence of a 620–650 Ma thermal event in Tanzania, but argue that continent–continent collision between East Africa and East Gondwana occurred at 550 Ma. Our argon data does not resolve this conflict, but it does suggest that if the northern stratoid granites were genetically related to post-collisional extension, then orogenic collapse took place shortly after the ~650 Ma main collision in north-central Madagascar. Subsequent metamorphism of the stratoid granites was localized in the region of the Antananarivo virgation zone at ~560 Ma (D_2) and along the borders of the Angavo Belt at ~550 Ma (D_3).

5. Conclusions

A paleomagnetic and $^{40}Ar/^{39}Ar$ study of the stratoid granites in central Madagascar revealed a complex history of thermo-chemical remagnetization during and following D_2 and D_3 deformational events in central Madagascar (< 550 Ma). A post-metamorphic cooling history for the D_2–D_3 affected region indicates that the magnetization was thermally reset at 521.4 ± 11.9 Ma. The paleomagnetic data are complex and some sites (particularly in areas to the north of the virgation zone) may have rotated about non-vertical axes. Our best estimate for the paleomagnetic pole for Madagascar is derived from sites within the virgation zone. This pole falls at 6.7°S, 352.6°E ($a_{95} = 14.2°$).

Our work in central Madagascar along with the recent work in southern Madagascar by De Wit et al. (2001) supports the notion that the emplacement of the northern stratoid granites followed

collisional events beginning at around 650 Ma. Structural and petrographic evidence supplied by Nédélec et al. (1994, 1995) further suggest that these granites were emplaced at mid-crustal levels in a region of extension. Our $^{40}Ar/^{39}Ar$ dating can be interpreted to reflect rapid cooling of the granites (25 °C/Ma) due to a combination of uplift and thermal decay. Subsequently, the stratoid granites in the Antananarivo virgation zone were reheated (750–800 °C) at slightly higher structural levels (3.5–3.6 kbars) resulting in a pervasive remagnetization. Post-deformational cooling in the Antananarivo region was fairly rapid (8 °C/Ma) during the 550–520 Ma interval and then slowed to about 2.5 °C/Ma at around 500 Ma assuming a simple exhumation path. Our interpretation of these data is that the deformation in the Angavo belt and the Antananarivo virgation zone resulted from intracratonic reworking of a crustal fragment already attached to East Africa by an earlier collisional phase (East African Orogen; Stern, 1994; Meert, in press). The later reworking may have resulted from renewed collisional tectonism during the final stages of Gondwana assembly further south.

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Appendix A: $^{40}Ar/^{39}Ar$ results

Name	CUM ^{39}K	% ATMOS	$^{37}Ca/^{39}K$	±	($^{40}Ar/^{36}Ar$)	±	$^{40}Ar/^{39}K$	±	AGE	±
STR-33a	Biotite									
N3a-50	5.38E−04	8.61E+01	3.66E−01	3.69E−01	3.43E+02	4.18E+01	1.31E+01	9.92E+00	2.39E+08	1.70E+08
N3a-100	3.06E−03	2.24E+01	2.12E−01	7.75E−02	1.32E+03	6.74E+02	1.60E+01	2.40E+00	2.89E+08	3.99E+07
N3a-150	8.16E−03	2.48E+01	2.64E−01	3.21E−02	1.19E+03	1.08E+02	2.45E+01	7.78E−01	4.25E+08	1.20E+07
N3a-200	2.53E−02	2.57E+00	3.93E−02	1.28E−02	1.15E+04	4.04E+03	2.89E+01	2.94E−01	4.90E+08	4.37E+06
N3a-250	*6.16E−02*	*9.36E−01*	*1.10E−02*	*6.19E−03*	*3.16E+04*	*1.95E+04*	*2.93E+01*	*1.94E−01*	*4.96E+08*	*2.87E+06*
N3a-300	*1.13E−01*	*3.04E−01*	–	–	*9.71E+04*	*1.12E+05*	*2.95E+01*	*1.24E−01*	*4.99E+08*	*1.84E+06*
N3a-400	*2.17E−01*	–	*2.54E−03*	*1.74E−03*	–	–	*2.97E+01*	*7.02E−02*	*5.03E+08*	*1.04E+06*
N3a-500	*3.25E−01*	*1.11E−01*	–	–	*2.65E+05*	*3.59E+05*	*2.97E+01*	*7.02E−02*	*5.03E+08*	*1.04E+06*
N3a-600	*4.17E−01*	*8.54E−02*	–	–	*3.46E+05*	*8.09E+05*	*2.99E+01*	*7.25E−02*	*5.05E+08*	*1.07E+06*
N3a-800	*5.97E−01*	*9.88E−02*	–	–	*2.99E+05*	*2.99E+05*	*2.96E+01*	*4.07E−02*	*5.02E+08*	*6.02E+05*
N3a-1200	*9.88E−01*	*6.05E−02*	*3.41E−03*	*5.59E−04*	*4.89E+05*	*4.40E+05*	*2.96E+01*	*1.84E−01*	*5.01E+08*	*2.72E+06*
N3a-1600	*9.95E−01*	–	*7.92E−02*	*3.05E−02*	–	–	*3.01E+01*	*7.48E−01*	*5.08E+08*	*1.10E+07*
N3a-2400	1.00E+00	7.69E+00	–	–	3.84E+03	1.46E+03	2.69E+01	8.95E−01	4.61E+08	1.35E+07
N3a-3000	1.00E+00	4.60E+02	–	–	6.43E+01	5.20E+01	–	6.36E+01	–	3.48E+09
STR-33b	Biotite									
N3b-50	9.86E−04	6.41E+01	1.00E−01	1.92E−01	4.61E+02	5.11E+01	1.11E+01	2.20E+00	2.05E+08	3.83E+07

Table 3 (*Continued*)

Name	CUM ^{39}K	% ATMOS	^{37}Ca/^{39}K	±	(^{40}Ar/^{36}Ar)	±	^{40}Ar/^{39}K	±	AGE	±
N3b-100	4.70E−03	7.45E+01	–	–	3.97E+02	1.56E+01	5.45E+00	6.26E−01	1.03E+08	1.15E+07
N3b-150	7.64E−03	4.64E+01	7.97E−02	5.99E−02	6.37E+02	3.74E+01	1.70E+01	8.73E−01	3.05E+08	1.44E+07
N3b-200	1.43E−02	3.84E+01	1.31E−02	2.74E−02	7.70E+02	2.10E+01	2.60E+01	4.63E−01	4.47E+08	7.05E+06
N3b-250	*3.12E−02*	*1.09E+01*	–	–	*2.72E+03*	*9.05E+01*	*2.86E+01*	*1.50E−01*	*4.86E+08*	*2.24E+06*
N3b-300	*6.37E−02*	*4.73E+00*	–	–	*6.22E+03*	*3.49E+02*	*2.95E+01*	*1.03E−01*	*4.99E+08*	*1.52E+06*
N3b-400	*1.42E−01*	*1.35E+00*	*3.38E−05*	*2.44E−03*	*2.19E+04*	*1.31E+03*	*2.96E+01*	*3.57E−02*	*5.02E+08*	*5.28E+05*
N3b-500	*2.36E−01*	*3.06E−01*	*2.19E−03*	*1.79E−03*	*9.65E+04*	*2.54E+04*	*2.99E+01*	*5.17E−02*	*5.06E+08*	*7.63E+05*
N3b-600	*3.27E−01*	*1.71E−01*	–	–	*1.73E+05*	*7.94E+04*	*2.99E+01*	*3.26E−02*	*5.06E+08*	*4.80E+05*
N3b-700	*4.10E−01*	*1.59E−01*	*2.71E−03*	*1.67E−03*	*1.85E+05*	*9.95E+04*	*2.99E+01*	*6.27E−02*	*5.06E+08*	*9.24E+05*
N3b-800	*4.90E−01*	*4.04E−01*	*1.97E−03*	*1.33E−03*	*7.31E+04*	*1.43E+04*	*2.99E+01*	*4.90E−02*	*5.06E+08*	*7.23E+05*
N3b-900	*5.81E−01*	*3.77E−01*	*4.91E−03*	*1.88E−03*	*7.83E+04*	*1.91E+04*	*2.97E+01*	*3.83E−02*	*5.03E+08*	*5.66E+05*
N3b-1000	*6.78E−01*	*2.49E−01*	*6.26E−03*	*1.46E−03*	*1.19E+05*	*4.73E+04*	*2.97E+01*	*4.51E−02*	*5.03E+08*	*6.66E+05*
N3b-1200	*8.47E−01*	*2.36E−01*	*2.84E−02*	*1.32E−03*	*1.25E+05*	*2.21E+04*	*2.96E+01*	*3.77E−02*	*5.02E+08*	*5.57E+05*
N3b-1600	*9.91E−01*	*8.07E−02*	*1.08E−02*	*1.78E−03*	*3.66E+05*	*3.73E+05*	*2.96E+01*	*4.39E−02*	*5.02E+08*	*6.49E+05*
N3b-2400	1.00E+00	5.71E−01	3.90E−01	2.25E−02	5.17E+04	9.64E+04	3.01E+01	3.41E−01	5.09E+08	5.02E+06
STR-54a	Biotite									
N2a-100	2.04E−03	9.93E+01	2.74E+00	4.67E−01	2.98E+02	1.49E+02	1.91E+01	1.28E+01	3.72E+06	2.49E+08
N2a-150	4.68E−03	4.19E+01	4.88E−01	1.50E−01	7.05E+02	1.91E+02	2.59E+01	5.11E+00	4.46E+08	7.78E+07
N2a-200	1.49E−02	2.18E+01	1.09E−02	3.94E−02	1.35E+03	2.03E+02	2.84E+01	1.25E+00	4.84E+08	1.87E+07
N2a-250	*3.53E−02*	*2.69E+00*	*1.66E−02*	*2.06E−02*	*9.99E+03*	*8.00E+03*	*3.02E+01*	*7.68E−01*	*5.11E+08*	*1.13E+07*
N2a-300	*6.04E−02*	–	–	–	–	–	*3.01E+01*	*5.05E−01*	*5.09E+08*	*7.44E+06*
N2a-400	*1.02E−01*	–	*1.98E−03*	*9.71E−03*	–	–	*2.97E+01*	*3.76E−01*	*5.03E+08*	*5.55E+06*
N2a-500	*1.64E−01*	*1.26E+00*	–	–	*2.35E+04*	*1.12E+04*	*2.94E+01*	*2.15E−01*	*4.99E+08*	*3.18E+06*
N2a-600	*2.58E−01*	*5.17E−02*	–	–	*5.72E+05*	*5.14E+06*	*2.97E+01*	*1.43E−01*	*5.03E+08*	*2.11E+06*
N2a-800	*5.08E−01*	*1.95E−01*	–	–	*1.52E+05*	*1.50E+05*	*2.95E+01*	*7.86E−02*	*5.00E+08*	*1.16E+06*
N2a-1200	*9.40E−01*	*2.60E−01*	–	–	*1.14E+05*	*3.93E+04*	*2.95E+01*	*6.03E−02*	*4.99E+08*	*8.93E+05*
N2a-1600	*9.90E−01*	*1.47E+00*	–	–	*2.02E+04*	*1.16E+04*	*2.92E+01*	*2.74E−01*	*4.95E+08*	*4.06E+06*
N2a-2400	*1.00E+00*	*3.42E+00*	–	–	*8.63E+03*	*1.10E+04*	*2.82E+01*	*1.31E+00*	*4.81E+08*	*1.96E+07*
N2a-3000	1.00E+00	–	–	–	–	–	2.83E+02	1.17E+03	2.53E+09	5.64E+09
STR-54b	Biotite									
N2b-50	1.53E−02	7.27E+01	2.84E−01	4.34E−02	4.07E+02	7.29E+00	3.50E+01	1.46E+01	5.80E+08	2.07E+08
N2b-100	1.67E−02	9.72E+01	9.97E−01	5.67E−01	3.04E+02	3.64E+00	3.48E+01	8.95E+00	5.77E+08	1.27E+08
N2b-150	2.10E−02	1.61E+01	1.32E+00	6.24E−01	1.83E+03	2.43E+03	3.13E+01	2.30E+00	5.26E+08	3.36E+07
N2b-200	3.54E−02	2.23E+01	2.46E−01	1.41E−01	1.33E+03	3.36E+02	3.10E+01	1.00E+00	5.22E+08	1.46E+07
N2b-250	*6.36E−02*	*1.74E+01*	*6.24E−02*	*4.44E−02*	*1.70E+03*	*2.57E+02*	*2.98E+01*	*4.19E−01*	*5.04E+08*	*6.19E+06*
N2b-300	*9.68E−02*	*9.92E+00*	*2.43E−02*	*2.64E−02*	*2.98E+03*	*3.41E+02*	*2.91E+01*	*3.69E−01*	*4.94E+08*	*5.48E+06*
N2b-400	*1.41E−01*	*6.22E+00*	*2.34E−02*	*1.95E−02*	*4.75E+03*	*8.76E+02*	*2.94E+01*	*2.47E−01*	*4.99E+08*	*3.65E+06*
N2b-500	*1.86E−01*	*2.22E+00*	*1.19E−02*	*1.83E−02*	*1.33E+04*	*4.37E+03*	*2.92E+01*	*2.47E−01*	*4.96E+08*	*3.67E+06*
STR-54b	Biotite									
N2b-600	*2.44E−01*	*4.33E+00*	*1.64E−02*	*1.58E−02*	*6.83E+03*	*1.07E+03*	*2.93E+00*	*1.93E−01*	*4.97E+08*	*2.86E+06*
N2b-700	*3.05E−01*	*3.79E+00*	–	*1.06E−02*	*7.80E+03*	*1.27E+03*	*2.98E+01*	*2.06E−01*	*5.04E+08*	*3.05E+06*

Table 3 (*Continued*)

Name	CUM ^{39}K	% ATMOS	^{37}Ca/^{39}K	±	(^{40}Ar/^{36}Ar)	±	^{40}Ar/^{39}K	±	AGE	±
N2b-800	*3.89E−01*	*2.43E+00*	*3.38E−02*	*3.19E−02*	*1.22E+04*	*2.40E+03*	*2.95E+01*	*1.29E−01*	*5.00E+08*	*1.91E+06*
N2b-900	*5.31E−01*	*2.00E+00*	*5.31E−03*	*5.71E−03*	*1.48E+04*	*2.50E+03*	*2.93E+01*	*1.02E−01*	*4.97E+08*	*1.51E+06*
N2b-1000	*6.94E−01*	*1.17E+00*	–	–	*2.52E+04*	*5.51E+03*	*2.96E+01*	*9.69E−02*	*5.02E+08*	*1.43E+06*
N2b-1200	*8.96E−01*	*5.35E−01*	–	–	*5.52E+04*	*2.06E+04*	*2.95E+01*	*7.54E−02*	*5.00E+08*	*1.12E+06*
N2b-1600	*9.48E−01*	*5.07E−01*	–	–	*5.83E+04*	*1.74E+04*	*2.95E+01*	*1.87E−01*	*4.99E+08*	*2.77E+06*
N2b-2400	*9.91E−01*	*1.84E−01*	–	–	*1.61E+05*	*4.76E+05*	*2.96E+01*	*2.13E−01*	*5.01E+08*	*3.16E+06*
N2b-3000	*1.00E+00*	*1.28E−01*	–	–	*2.31E+05*	*1.15E+06*	*2.90E+01*	*9.21E−01*	*4.93E+08*	*1.37E+07*
STR-84a	Biotite									
N0a-50	2.18E−03	9.46E+01	2.06E−01	1.03E−01	3.12E+02	4.43E+01	1.92E+00	4.80E+00	3.71E+07	9.19E+07
N0a-100	6.50E−03	7.97E+01	1.23E−01	5.80E−02	3.71E+02	2.57E+01	7.50E+00	2.05E+00	1.41E+08	3.71E+07
N0a-150	1.44E−02	2.76E+01	2.48E−02	3.35E−02	1.07E+03	1.18E+02	2.19E+01	9.36E−01	3.84E+08	1.48E+07
N0a-200	3.61E−02	9.03E+00	5.77E−02	1.05E−02	3.27E+03	3.58E+02	2.73E+01	3.17E−01	4.68E+08	4.78E+06
N0a-250	7.92E−02	4.63E+00	9.32E−03	5.13E−03	6.38E+03	6.25E+02	2.79E+01	1.76E−01	4.77E+08	2.64E+06
N0a-300	1.37E−01	2.92E+00	5.57E−05	3.94E−03	1.01E+04	1.36E+03	2.83E+01	1.71E−01	4.82E+08	2.56E+06
N0a-400	2.35E−01	3.76E+00	2.00E−03	1.81E−03	7.85E+03	4.33E+02	2.81E+01	7.85E−02	4.79E+08	1.18E+06
N0a-500	*3.28E−01*	*1.98E+00*	*3.04E−03*	*2.48E−03*	*1.49E+04*	*1.94E+03*	*2.88E+01*	*1.03E−01*	*4.90E+08*	*1.53E+06*
N0a-600	*4.08E−01*	*1.32E+00*	–	–	*2.23E+04*	*5.03E+03*	*2.92E+01*	*1.24E−01*	*4.96E+08*	*1.84E+06*
N0a-800	*5.57E−01*	*1.22E+00*	–	–	*2.43E+04*	*3.60E+03*	*2.91E+01*	*6.59E−02*	*4.95E+08*	*9.79E+05*
N0a-1200	*9.26E−01*	*3.52E−01*	*2.18E−03*	*7.68E−04*	*8.40E+04*	*1.83E+04*	*2.91E+01*	*3.01E−02*	*4.94E+08*	*4.47E+05*
N0a-1600	*9.87E−01*	–	*3.16E−03*	*4.15E−03*	–	–	*2.93E+01*	*1.37E−01*	*4.98E+08*	*2.03E+06*
N0a-2400	1.00E+00	–	–	1.96E−02	–	–	3.05E+01	4.81E−01	5.15E+08	7.06E+06
N0a-3000	1.00E+00	–	8.26E−01	1.25E+00	–	–	1.85E+02	3.96E+01	1.98E+09	2.58E+08
STR-84b	Biotite									
N0b-50	5.86E−04	5.29E+01	1.23E−01	6.19E−01	5.59E+02	4.98E+02	1.37E+01	1.37E+01	2.50E+08	2.33E+08
N0b-100	1.51E−03	9.03E+01	–	5.47E−01	3.27E+02	1.22E+02	3.45E+00	1.20E+01	6.62E+07	2.25E+08
N0b-150	2.73E−03	6.72E+01	2.04E−01	–	4.40E+02	1.80E+02	1.00E+01	8.42E+00	1.86E+08	1.48E+08
N0b-200	4.29E−03	4.84E+01	–	2.06E−01	6.11E+02	1.72E+02	2.13E+01	5.64E+00	3.74E+08	8.96E+07
N0b-250	7.29E−03	2.13E+01	2.60E−02	1.22E−01	1.39E+03	7.41E+02	2.32E+01	3.38E+00	4.04E+08	5.27E+07
N0b-300	1.44E−02	9.08E+00	–	–	3.25E+03	8.07E+02	2.82E+01	8.10E−01	4.81E+08	1.21E+07
N0b-400	4.11E−02	5.34E+00	–	–	5.53E+03	7.48E+02	2.81E+01	2.26E−01	4.79E+08	3.39E+06
N0b-500	8.76E−02	2.62E+00	–	–	1.13E+04	2.34E+03	2.86E+01	2.00E−01	4.88E+08	2.98E+06
N0b-600	1.46E−01	1.84E+00	–	–	1.61E+04	3.35E+03	2.88E+01	1.43E−01	4.90E+08	2.14E+06
N0b-700	*2.04E−01*	*1.19E+00*	–	–	*2.47E+04*	*6.74E+03*	*2.91E+01*	*1.29E−01*	*4.94E+08*	*1.91E+06*
N0b-800	*2.65E−01*	*7.97E−01*	–	–	*3.71E+04*	*1.36E+04*	*2.93E+01*	*1.04E−01*	*4.97E+08*	*1.54E+06*
N0b-900	*3.24E−01*	*6.49E−01*	*6.31E−03*	*5.65E−03*	*4.55E+04*	*2.28E+04*	*2.95E+01*	*1.30E−01*	*5.00E+08*	*1.92E+06*
N0b-1000	*3.89E−01*	*3.65E−01*	–	–	*8.10E+04*	*5.96E+04*	*2.94E+01*	*1.37E−01*	*4.99E+08*	*2.02E+06*
N0b-1200	*5.35E−01*	*5.96E−01*	*3.13E−03*	*2.95E−03*	*4.95E+04*	*1.11E+04*	*2.91E+01*	*8.32E−02*	*4.94E+08*	*1.24E+06*
N0b-1600	*9.55E−01*	*2.55E−01*	*1.12E−03*	*8.30E−04*	*1.16E+05*	*2.45E+04*	*2.92E+01*	*3.00E−02*	*4.96E+08*	*4.46E+05*
N0b-2400	9.99E−01	–	–	–	–	–	2.93E+01	2.15E−01	4.98E+08	3.19E+06
N0b-3000	1.00E+00	1.87E+01	–	–	1.58E+03	1.45E+03	2.50E+01	5.34E+00	4.32E+08	8.22E+07

Table 3 (*Continued*)

Name	CUM ^{39}K	% ATMOS	^{37}Ca/^{39}K	±	(^{40}Ar/^{36}Ar)	±	^{40}Ar/^{39}K	±	AGE	±
STR-57	Hornblende									
N4a-100	2.27E−03	1.93E+01	7.32E+00	1.20E+00	1.53E+03	5.18E+02	2.47E+02	2.51E+01	2.34E+09	1.33E+08
N4a-200	5.12E−03	1.05E+01	1.74E+00	9.74E−01	2.82E+03	5.64E+03	7.27E+01	1.75E+01	1.05E+09	1.91E+08
N4a-275	7.91E−03	3.28E+01	2.01E+00	9.99E−01	9.02E+02	5.84E+02	4.73E+01	1.53E+01	7.45E+08	1.97E+08
N4a-325	1.03E−02	3.48E+01	6.57E−01	1.11E+00	8.49E+02	1.11E+03	2.35E+01	1.66E+01	4.08E+08	2.58E+08
N4a-450	2.93E−02	3.12E+00	2.86E+00	1.95E−01	9.46E+03	2.02E+04	3.50E+01	2.45E+00	5.78E+08	3.46E+07
N4a-500	*5.34E−02*	*8.14E+00*	*3.04E+00*	*1.12E−01*	*3.63E+03*	*2.45E+03*	*3.07E+01*	*1.85E+00*	*5.16E+08*	*2.71E+07*
N4a-550	*7.78E−02*	*5.90E+00*	*3.27E+00*	*1.17E−01*	*5.01E+03*	*4.53E+03*	*2.98E+01*	*1.70E+00*	*5.03E+08*	*2.51E+07*
N4a-600	*1.29E−01*	*1.04E+00*	*3.35E+00*	*7.29E−02*	*2.85E+04*	*8.03E+04*	*3.01E+01*	*9.20E−01*	*5.08E+08*	*1.35E+07*
N4a-625	*1.71E−01*	*2.80E+00*	*3.39E+00*	*8.38E−02*	*1.05E+04*	*1.17E+04*	*3.02E+01*	*1.01E+00*	*5.10E+08*	*1.49E+07*
N4a-650	*2.29E−01*	*8.97E−01*	*3.58E+00*	*8.00E−02*	*3.30E+04*	*8.70E+04*	*3.01E+01*	*7.29E−01*	*5.08E+08*	*1.07E+07*
N4a-675	*3.43E−01*	–	*3.47E+00*	*4.82E−02*	–	–	*3.06E+01*	*2.92E−01*	*5.15E+08*	*4.28E+06*
N4a-700	*4.13E−01*	–	*3.41E+00*	*4.81E−02*	–	–	*3.10E+01*	*5.40E−01*	*5.22E+08*	*7.88E+06*
STR-57	Hornblende									
N4a-725	*5.28E−01*	–	–	–	*3.94E−02*	–	*3.04E+01*	*4.23E−01*	*5.12E+08*	*6.21E+06*
*N4a-750**	*5.31E−01*	–	–	–	*7.71E−01*	–	*3.79E+01*	*1.32E+01*	*6.19E+08*	*1.82E+08*
*N4a-775**	*5.35E−01*	–	–	–	*4.07E−01*	–	*2.80E+01*	*7.27E+00*	*4.77E+08*	*1.09E+08*
*N4a-800**	*5.42E−01*	–	–	–	*3.15E−01*	–	*3.33E+01*	*4.20E+00*	*5.55E+08*	*6.01E+07*
N4a-850	*5.62E−01*	*1.22E+00*	*1.22E+00*	*3.34E+00*	*1.47E−01*	*2.41E+04*	*3.03E+01*	*1.39E+00*	*5.11E+08*	*2.03E+07*
N4a-900	*5.81E−01*	*6.88E−01*	*6.88E−01*	*3.07E+00*	*1.11E−01*	*4.29E+04*	*3.04E+01*	*2.19E+00*	*5.12E+08*	*3.21E+07*
N4a-1000	*6.71E−01*	–	–	–	*4.83E−02*	–	*3.05E+01*	*3.20E−01*	*5.14E+08*	*4.69E+06*
N4a-1200	*8.25E−01*	*6.20E−02*	*6.20E−02*	*3.49E+00*	*3.27E−02*	*4.76E+05*	*3.07E+01*	*1.79E−01*	*5.17E+08*	*2.62E+06*
N4a-1400	*9.09E−01*	*1.19E−01*	*1.19E−01*	*3.83E+00*	*4.15E−02*	*2.48E+05*	*3.11E+01*	*4.62E−01*	*5.22E+08*	*6.73E+06*
N4a-3000	1.00E+00	6.81E−01	6.81E−01	3.58E+00	2.96E−02	4.34E+04	3.44E+01	3.75E−01	5.70E+08	5.33E+06
STR-84	K-spar									
M9a-50	1.85E−04	3.18E+01	–	–	9.29E+02	3.86E+02	4.22E+01	8.42E+00	6.81E+08	1.13E+08
M9a-100	6.47E−04	1.39E−01	–	–	2.13E−03	1.56E+03	3.05E+01	3.68E+00	5.15E+08	5.42E+07
M9a-125	1.30E−03	–	–	–	–	–	2.54E+01	2.41E+00	4.39E+08	3.70E+07
M9a-150	2.77E−03	1.33E+01	–	–	2.23E+03	7.46E+02	2.22E+01	1.15E+00	3.89E+08	1.81E+07
M9a-175	5.51E−03	3.85E+00	–	–	7.68E+03	5.73E+03	2.18E+01	6.58E−01	3.83E+08	1.04E+07
M9a-200	1.01E−02	4.36E+00	1.36E−03	1.91E−02	6.78E+03	2.54E+03	2.26E+01	4.05E−01	3.96E+08	6.37E+06
M9a-225	1.68E−02	9.13E−01	7.40E−03	1.59E−02	3.24E+04	4.57E+04	2.29E+01	3.11E−01	4.00E+08	4.87E+06
M9a-250	2.64E−02	–	7.59E−03	1.14E−02	–	–	2.36E+01	2.02E−01	4.11E+08	3.14E+06
M9a-275	3.59E−02	–	1.31E−02	1.24E−02	–	–	2.38E+01	2.35E−01	4.14E+08	3.65E+06
M9a-300	4.79E−02	–	1.35E−02	8.69E−03	–	–	2.39E+01	1.64E−01	4.16E+08	2.55E+06
M9a-325	6.21E−02	–	2.89E−02	5.80E−03	–	–	2.42E+01	1.49E−01	4.21E+08	2.30E+06
M9a-350	7.78E−02	–	3.32E−02	4.72E−03	–	–	2.41E+01	1.26E−01	4.19E+08	1.95E+06
M9a-375	9.51E−02	–	2.76E−02	6.58E−03	–	–	2.41E+01	1.21E−01	4.19E+08	1.88E+06
M9a-400	1.13E−01	–	3.74E−02	6.54E−03	–	–	2.45E+01	9.71E−02	4.25E+08	1.50E+06
M9a-425	1.33E−01	–	3.38E−02	3.71E−03	–	–	2.45E+01	1.19E−01	4.26E+08	1.84E+06
M9a-450	1.52E−01	–	3.45E−02	5.14E−03	–	–	2.46E+01	9.28E−02	4.26E+08	1.43E+06
M9a-475	1.73E−01	1.20E−01	2.98E−02	6.54E−03	2.46E+05	4.35E+05	2.46E+01	8.37E−02	4.27E+08	1.29E+06
M9a-500	1.93E−01	–	2.22E−02	6.08E−03	–	–	2.47E+01	8.26E−02	4.28E+08	1.28E+06

Table 3 (*Continued*)

Name	CUM ^{39}K	% ATMOS	^{37}Ca/^{39}K	±	(^{40}Ar/^{36}Ar)	±	^{40}Ar/^{39}K	±	AGE	±
M9a-525	2.12E−01	6.24E−01	2.43E−02	7.28E−03	4.74E+04	2.07E+04	2.48E+01	8.99E−02	4.29E+08	1.39E+06
M9a-550	2.31E−01	3.93E−01	2.49E−02	7.25E−03	7.51E+04	6.00E+04	2.48E+01	9.18E−02	4.29E+08	1.42E+06
M9a-575	2.50E−01	3.88E−01	1.37E−02	7.56E−03	7.61E+04	3.79E+04	2.49E+01	8.73E−02	4.32E+08	1.35E+06
M9a-600	2.67E−01	4.35E−01	1.48E−02	8.51E−03	6.79E+04	2.73E+04	2.49E+01	7.27E−02	4.31E+08	1.12E+06
M9a-625	2.83E−01	4.76E−01	2.72E−02	7.05E−03	6.21E+04	4.63E+04	2.52E+01	1.11E−01	4.36E+08	1.70E+06
M9a-650	2.99E−01	3.34E−02	1.42E−02	6.25E−03	8.85E+05	8.94E+06	2.53E+01	1.02E−01	4.38E+08	1.57E+06
M9a-675	3.14E−01	7.60E−02	2.89E−02	5.49E−03	3.89E+05	1.94E+06	2.57E+01	1.17E−01	4.44E+08	1.80E+06
M9a-700	3.28E−01	4.91E−02	2.07E−02	5.17E−03	6.02E+05	3.72E+06	2.58E+01	1.02E−01	4.45E+08	1.56E+06
M9a-725	3.42E−01	–	1.75E−02	4.07E−03	–	–	2.58E+01	1.13E−01	4.45E+08	1.72E+06
M9a-750	3.55E−01	1.07E+00	1.77E−02	7.46E−03	2.77E+04	1.22E+04	2.56E+01	1.59E−01	4.42E+08	2.44E+06
M9a-775	3.68E−01	3.70E−01	1.24E−02	9.86E−03	7.98E+04	1.22E+05	2.60E+01	1.57E−01	4.49E+08	2.40E+06
M9a-800	3.81E−01	1.69E−01	1.91E−02	1.06E−02	1.75E+05	3.46E+05	2.60E+01	1.03E−01	4.49E+08	1.58E+06
M9a-825	3.94E−01	1.06E−01	4.25E−03	9.36E−03	2.80E+05	6.00E+05	2.63E+01	7.65E−02	4.53E+08	1.16E+06
M9a-850	4.06E−01	–	2.38E−02	1.04E−02	–	–	2.63E+01	2.01E−01	4.53E+08	3.06E+06
M9a-875	4.19E−01	2.88E−01	1.02E−02	1.33E−02	1.02E+05	1.39E+05	2.62E+01	1.35E−01	4.52E+08	2.06E+06
M9a-900	4.31E−01	–	9.07E−03	1.06E−02	–	–	2.62E+01	1.10E−01	4.51E+08	1.68E+06
M9a-925	4.43E−01	3.55E−01	2.28E−02	8.19E−03	8.33E+04	1.15E+05	2.63E+01	1.62E−01	4.53E+08	2.47E+06
M9a-950	4.56E−01	4.78E−01	1.56E−02	8.47E−03	6.18E+04	6.33E+04	2.61E+01	1.45E−01	4.50E+08	2.21E+06
M9a-975	4.69E−01	2.30E−01	2.08E−02	7.92E−03	1.28E+05	2.11E+05	2.62E+01	1.06E−01	4.52E+08	1.62E+06
M9a-1000	4.83E−01	3.45E−01	2.03E−02	8.45E−03	8.56E+04	1.10E+05	2.63E+01	1.39E−01	4.53E+08	2.12E+06
M9a-1025	4.97E−01	1.78E−01	1.77E−02	8.19E−03	1.66E+05	4.62E+05	2.62E+01	1.46E−01	4.51E+08	2.23E+06
M9a-1050	5.12E−01	1.73E−01	1.24E−02	1.10E−02	1.71E+05	6.10E+05	2.62E+01	1.70E−01	4.51E+08	2.59E+06
M9a-1075	5.27E−01	4.98E−01	1.79E−02	7.49E−03	5.94E+04	4.66E+04	2.62E+01	1.19E−01	4.51E+08	1.81E+06
M9a-1100	5.43E−01	1.12E−01	1.58E−02	8.20E−03	2.64E+05	9.69E+05	2.62E+01	1.28E−01	4.52E+08	1.95E+06
M9a-1125	5.58E−01	4.38E−01	7.40E−03	6.26E−03	6.75E+04	4.41E+04	2.62E+01	9.32E−02	4.51E+08	1.42E+06
M9a-1150	5.74E−01	6.77E−01	1.49E−02	7.58E−03	4.36E+04	2.26E+04	2.60E+01	1.26E−01	4.49E+08	1.93E+06
M9a-1175	5.50E−01	5.19E−01	1.53E−02	6.69E−03	5.69E+04	3.68E+04	2.61E+01	1.18E−01	4.50E+08	1.81E+06
M9a-1200	6.06E−01	2.52E−01	3.65E−03	7.31E−03	1.17E+05	1.48E+05	2.63E+01	1.19E−01	4.52E+08	1.82E+06
M9a-1225	6.22E−01	7.22E−02	3.93E−03	6.44E−03	4.09E+05	2.42E+06	2.63E+01	1.18E−01	4.52E+08	1.80E+06
M9a-1250	6.37E−01	4.77E−01	6.43E−03	7.11E−03	6.19E+04	5.05E+04	2.61E+01	1.16E−01	4.50E+08	1.77E+06
M9a-1275	6.53E−01	4.36E−02	3.01E−04	6.14E−03	6.78E+05	6.40E+06	2.62E+01	1.16E−01	4.51E+08	1.78E+06
M9a-1300	6.67E−01	6.36E−01	3.72E−03	7.03E−03	4.65E+04	2.86E+04	2.62E+01	1.42E−01	4.52E+08	2.17E+06
M9a-1350	6.81E−01	8.83E−01	7.47E−03	9.98E−03	3.35E+04	1.28E+04	2.58E+01	9.47E−02	4.45E+08	1.45E+06
M9a-1400	6.96E−01	2.45E−01	3.44E−03	6.37E−03	1.21E+05	2.44E+05	2.62E+01	1.74E−01	4.51E+08	2.65E+06
M9a-1500	7.17E−01	2.93E−01	6.89E−03	5.21E−03	1.01E+05	9.06E+04	2.62E+01	9.50E−02	4.51E+08	1.45E+06
M9a-1600	7.42E−01	3.99E−01	1.07E−02	4.62E−03	7.41E+04	3.14E+04	2.63E+01	7.91E−02	4.54E+08	1.20E+06
M9a-1700	7.69E−01	1.00E+00	1.20E−02	4.44E−03	2.95E+04	4.09E+03	2.61E+01	5.27E−02	4.49E+08	8.04E+05
M9a-1800	7.95E−01	2.06E−01	5.28E−03	3.64E−03	1.43E+05	1.76E+05	2.63E+01	7.42E−02	4.53E+08	1.13E+06
M9a-1900	8.19E−01	2.91E−02	7.46E−03	4.78E−03	1.02E+06	7.92E+06	2.65E+01	9.00E−02	4.56E+08	1.37E+06
M9a-2000	8.41E−01	5.38E−01	1.18E−02	4.60E−03	5.49E+04	3.60E+04	2.65E+01	1.13E−01	4.56E+08	1.71E+06
M9a-2200	8.68E−01	4.67E−01	1.81E−02	5.20E−03	6.33E+04	3.88E+04	2.67E+01	9.12E−02	4.59E+08	1.38E+06
M9a-2400	8.96E−01	3.44E−01	1.08E−02	3.62E−03	8.59E+04	7.52E+04	2.67E+01	9.25E−02	4.60E+08	1.40E+06
M9a-2600	9.22E−01	7.54E−01	1.92E−02	4.99E−03	3.92E+04	1.62E+04	2.67E+01	9.20E−02	4.60E+08	1.40E+06
M9a-2800	9.43E−01	2.88E−01	9.30E−03	6.14E−03	1.03E+05	1.43E+05	2.69E+01	1.19E−01	4.62E+08	1.80E+06

Table 3 (*Continued*)

Name	CUM ^{39}K	% ATMOS	^{37}Ca/^{39}K	±	(^{40}Ar/^{36}Ar)	±	^{40}Ar/^{39}K	±	AGE	±
M9a-3000	9.59E−01	6.71E−01	2.43E−02	5.47E−03	4.40E+04	3.36E+04	2.71E+01	1.49E−01	4.65E+08	2.26E+06
M9a-3200	9.71E−01	3.60E−01	3.72E−02	1.07E−02	8.20E+04	9.11E+04	2.73E+01	1.29E−01	4.68E+08	1.95E+06
M9a-3400	9.81E−01	1.73E−01	1.67E−02	1.12E−02	1.71E+05	4.30E+05	2.70E+01	1.37E−01	4.64E+08	2.07E+06
M9a-3600	9.88E−01	8.93E−02	3.02E−02	1.43E−02	3.31E+05	2.44E+06	2.74E+01	2.32E−01	4.69E+08	3.50E+06
M9a-3800	9.96E−01	4.24E−01	1.90E−02	1.85E−02	6.97E+04	9.22E+04	2.70E+01	1.64E−01	4.64E+08	2.49E+06
M9a-4000	1.00E+00	1.70E+00	2.72E−02	2.20E−02	1.73E+04	9.55E+03	2.68E+01	2.85E−01	4.61E+08	4.32E+06
STR-11a	Biotite									
wla-50	1.92E−03	6.54E+01	4.85E−02	7.81E−03	4.52E+02	2.71E+02	8.76E+00	1.00E+00	1.43E+08	1.58E+07
wla-75	3.70E−03	5.25E+01	7.86E−02	8.72E−03	5.64E+02	5.21E+01	9.17E+00	9.40E−01	1.50E+08	1.47E+07
wla-100	5.47E−03	3.41E+01	9.24E−02	6.37E−03	8.66E+02	1.15E+02	1.93E+01	1.35E+00	3.01E+08	1.94E+07
wla-125	8.35E−03	2.27E+01	4.78E−02	8.37E−03	1.30E+03	8.86E+01	3.10E+01	6.82E−01	4.62E+08	8.97E+06
wla-150	1.40E−02	1.28E+03	4.10E−02	2.17E−03	2.31E+03	1.69E+02	3.81E+01	4.37E−01	5.53E+08	5.47E+06
wla-175	2.37E−02	6.48E+00	4.79E−03	1.65E−03	4.56E+03	3.39E+02	4.17E+01	3.02E−01	5.98E+08	3.68E+06
wla-200	*3.67E−02*	*2.25E+00*	*2.56E−03*	*1.08E−03*	*1.51E+04*	*2.80E+03*	*4.27E+01*	*2.86E−01*	*6.10E+08*	*3.47E+06*
wla-225	*5.19E−02*	*1.77E+00*	*4.48E−03*	*1.32E−03*	*1.67E+04*	*3.61E+03*	*4.31E+01*	*5.06E−01*	*6.14E+08*	*6.12E+06*
wla-250	*7.10E−02*	*1.66E+00*	*4.17E−03*	*7.26E−04*	*1.78E+04*	*2.41E+03*	*4.28E+01*	*1.44E−01*	*6.11E+08*	*1.75E+06*
wla-275	*9.19E−02*	*1.29E+00*	*4.63E−03*	*5.10E−04*	*2.29E+04*	*3.98E+03*	*4.30E+01*	*1.78E−01*	*6.14E+08*	*2.16E+06*
wla-300	*1.15E−01*	*1.27E+00*	*3.48E−03*	*5.56E−04*	*2.32E+04*	*4.07E+03*	*4.29E+01*	*1.01E−01*	*6.13E+08*	*1.22E+06*
wla-325	*1.39E−01*	*1.09E+00*	*4.40E−03*	*7.12E−04*	*2.71E+04*	*4.15E+03*	*4.29E+01*	*1.16E−01*	*6.12E+08*	*1.40E+06*
wla-350	*1.63E−01*	*5.86E−01*	*2.72E−03*	*5.52E−04*	*5.04E+04*	*1.70E+04*	*4.30E+01*	*1.42E−01*	*6.13E+08*	*1.71E+06*
wla-375	*1.87E−01*	*1.02E+00*	*3.79E−03*	*5.80E−04*	*2.88E+04*	*3.54E+03*	*4.30E+01*	*1.06E−01*	*6.13E+08*	*1.28E+06*
wla-400	*2.12E−01*	*5.97E−01*	*3.16E−03*	*4.86E−04*	*4.95E+04*	*1.90E+04*	*4.31E+01*	*1.33E−01*	*6.14E+08*	*1.61E+06*
wla-425	*2.35E−01*	*4.13E−01*	*7.14E−04*	*6.98E−04*	*7.15E+04*	*3.12E+04*	*4.33E+01*	*1.50E−01*	*6.17E+08*	*1.81E+06*
wla-450	*2.58E−01*	*3.34E−01*	*6.97E−04*	*6.71E−04*	*8.85E+04*	*4.10E+04*	*4.29E+01*	*1.27E−01*	*6.13E+08*	*1.53E+06*
wla-475	*2.82E−01*	*5.54E−01*	*1.64E−03*	*6.14E−04*	*5.33E+04*	*1.42E+04*	*4.29E+01*	*1.28E−01*	*6.12E+08*	*1.54E+06*
wla-500	*3.06E−01*	*1.92E−01*	*1.88E−03*	*7.39E−04*	*1.54E+05*	*1.01E+05*	*4.30E+01*	*1.04E−01*	*6.13E+08*	*1.25E+06*
wla-525	*3.32E−01*	*3.21E−01*	*1.53E−03*	*5.76E−04*	*9.22E+05*	*5.50E+06*	*4.30E+01*	*1.30E−01*	*6.14E+08*	*1.57E+06*
wla-550	*3.54E−01*	*2.53E−01*	*7.37E−04*	*8.02E−04*	*1.17E+05*	*9.38E+04*	*4.25E+01*	*1.83E−01*	*6.08E+08*	*2.22E+06*
wla-575	*3.75E−01*	*2.92E−01*	*9.04E−04*	*8.53E−04*	*1.01E+05*	*9.16E+04*	*4.30E+01*	*2.33E−01*	*6.13E+08*	*2.82E+06*
wla-600	*3.97E−01*	*2.07E−01*	–	–	*1.43E+05*	*8.53E+04*	*4.28E+01*	*1.09E−01*	*6.11E+08*	*1.33E+06*
wla-625	*4.19E−01*	*1.70E−01*	*1.64E−03*	*5.65E−04*	*1.74E+05*	*1.87E+05*	*4.31E+01*	*1.29E−01*	*6.14E+08*	*1.55E+06*
wla-650	*4.40E−01*	–	*1.24E−03*	*6.81E−04*	–	–	*4.32E+01*	*2.13E−01*	*6.16E+08*	*2.58E+06*
wla-675	*4.64E−01*	–	*1.10E−03*	*5.10E−04*	–	–	*4.32E+01*	*1.31E−01*	*6.16E+08*	*1.58E+06*
wla-700	*4.87E−01*	*1.28E−01*	*1.81E−03*	*6.68E−04*	*2.32E+05*	*3.96E+05*	*4.30E+01*	*1.23E−01*	*6.14E+08*	*1.49E+06*
wla-725	*5.13E−01*	–	*1.25E−03*	*4.61E−04*	–	–	*4.31E+01*	*1.24E−01*	*6.14E+08*	*1.50E+06*
wla-750	*5.38E−01*	*9.79E−02*	*1.64E−03*	*6.31E−04*	*3.02E+05*	*7.17E+05*	*4.29E+01*	*1.75E−01*	*6.12E+08*	*2.12E+06*
wla-775	*5.65E−01*	*2.61E−01*	*9.12E−04*	*6.41E−04*	*1.13E+05*	*8.13E+04*	*4.30E+01*	*1.29E−01*	*6.13E+08*	*1.56E+06*
wla-800	*5.90E−01*	*3.24E−01*	*1.48E−03*	*5.22E−04*	*9.13E+04*	*5.14E+04*	*4.27E+01*	*1.54E−01*	*6.10E+08*	*1.87E+06*
wlb-900	*4.06E−01*	*7.05E−03*	*3.98E−03*	*6.97E−04*	*4.19E+06*	*1.06E+08*	*4.24E+01*	*1.39E−01*	*6.07E+08*	*1.69E+06*
wlb-925	*4.22E−01*	*9.54E−02*	*1.59E−03*	*7.03E−04*	*3.10E+05*	*6.91E+05*	*4.26E+01*	*1.21E−01*	*6.09E+08*	*1.47E+06*
wlb-950	*4.38E−01*	*3.88E−01*	*1.67E−03*	*9.60E−04*	*7.62E+04*	*3.29E+04*	*4.23E+01*	*1.18E−01*	*6.05E+08*	*1.43E+06*
wlb-975	*4.54E−01*	*8.41E−02*	*2.42E−03*	*8.13E−04*	*3.51E+05*	*4.97E+05*	*4.26E+01*	*1.43E−01*	*6.08E+08*	*1.73E+06*
wlb-1000	*4.74E−01*	*3.76E−01*	*3.58E−03*	*7.57E−04*	*7.85E+04*	*4.51E+04*	*4.26E+01*	*1.35E−01*	*6.09E+08*	*1.63E+06*

Table 3 (*Continued*)

Name	CUM ^{39}K	% ATMOS	^{37}Ca/^{39}K	±	(^{40}Ar/^{36}Ar)	±	^{40}Ar/^{39}K	±	AGE	±
wlb-1050	*4.98E−01*	*3.78E−01*	*3.84E−03*	*5.79E−04*	*7.83E+04*	*2.35E+04*	*4.26E+01*	*8.32E−02*	*6.08E+08*	*1.01E+06*
wlb-1100	*5.27E−01*	*2.78E−01*	*3.92E−03*	*4.38E−04*	*1.06E+05*	*2.87E+04*	*4.24E+01*	*8.06E−02*	*6.07E+08*	*9.79E+05*
wlb-1150	*5.59E−01*	*2.74E−01*	*3.49E−03*	*4.89E−04*	*1.08E+05*	*4.32E+04*	*4.25E+01*	*7.87E−02*	*6.08E+08*	*9.55E+05*
wlb-1200	*5.93E−01*	*1.72E−01*	*3.15E−03*	*3.94E−04*	*1.72E+05*	*1.07E+05*	*4.26E+01*	*8.31E−02*	*6.09E+08*	*1.01E+06*
wlb-1300	*6.47E−01*	*2.49E−01*	*3.38E−03*	*1.97E−04*	*1.19E+05*	*3.52E+04*	*4.26E+01*	*5.77E−02*	*6.08E+08*	*7.00E+05*
wlb-1350	*6.94E−01*	*2.65E−01*	*2.88E−03*	*3.05E−04*	*1.11E+05*	*3.46E+04*	*4.24E+01*	*5.84E−02*	*6.06E+08*	*7.09E+05*
wlb-1375	*7.30E−01*	*2.91E−01*	*3.46E−03*	*3.67E−04*	*1.02E+05*	*4.42E+04*	*4.25E+01*	*8.46E−02*	*6.08E+08*	*1.03E+06*
wlb-1400	*7.58E−01*	*1.87E−01*	*3.40E−03*	*3.77E−04*	*1.58E+05*	*7.68E+04*	*4.25E+01*	*1.49E−01*	*6.08E+08*	*1.81E+06*
wlb-1425	*7.83E−01*	*2.35E−01*	*3.67E−03*	*3.45E−04*	*1.26E+05*	*6.22E+04*	*4.27E+01*	*9.38E−02*	*6.10E+08*	*1.14E+06*
wlb-1450	*8.07E−01*	*2.79E−01*	*4.03E−03*	*4.28E−04*	*1.06E+05*	*4.45E+04*	*4.27E+01*	*9.67E−02*	*6.10E+08*	*1.17E+06*
wlb-1475	*8.29E−01*	*1.85E−01*	*4.68E−03*	*4.86E−04*	*1.60E+05*	*1.23E+05*	*4.26E+01*	*8.57E−02*	*6.09E+08*	*1.04E+06*
wlb-1550	*8.47E−01*	*1.80E−01*	*4.30E−03*	*5.08E−04*	*1.64E+05*	*1.27E+05*	*4.26E+01*	*1.12E−01*	*6.09E+08*	*1.36E+06*
wlb-1600	*8.68E−01*	–	*3.16E−03*	*6.01E−04*	–	–	*4.29E+01*	*1.10E−01*	*6.13E+08*	*1.33E+06*
wlb-1700	*8.95E−01*	*1.28E−01*	*3.66E−03*	*4.09E−04*	*2.30E+05*	*2.84E+05*	*4.27E+01*	*8.56E−02*	*6.10E+08*	*1.04E+06*
wlb-1800	*9.28E−01*	*1.57E−01*	*3.42E−03*	*2.76E−04*	*1.88E+05*	*1.70E+05*	*4.30E+01*	*7.59E−02*	*6.14E+08*	*9.18E+05*
wlb-2000	*9.76E−01*	*1.04E−01*	*1.88E−03*	*2.54E−04*	*2.84E+05*	*2.50E+05*	*4.31E+01*	*6.76E−02*	*6.15E+08*	*8.17E+05*
wlb-2400	*9.95E−01*	*1.41E−01*	*4.08E−03*	*8.94E−04*	*2.10E+05*	*3.33E+05*	*4.32E+01*	*1.32E−01*	*6.15E+08*	*1.60E+06*
wlb-2800	9.99E−01	8.66E−01	8.07E−03	2.26E−03	3.41E+04	4.10E+04	4.42E+01	5.03E−01	6.27E+08	6.04E+06
wlb-3600	1.00E+00	1.02E+00	–	–	2.88E+04	1.23E+05	4.48E+01	2.09E+00	6.35E+08	2.50E+07

Below detectable limits are denoted by '–'. Column 1 gives simple packet number and laser power in milliwatt. Italics represents steps used to calculate plateau, integrated and inverse isochron ages reported in Table 3. *These low percentage gas release steps (with high errors) were not plotted in the figure although they were used in the calculation of the plateau age for STR-57.

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